Draft version October 12, 2026
Typeset using LaTeX **twocolumn** style in AASTeX7.0.1

# A grid of fast-rotating, chemically-homogeneous, supernova and/or long-GRB progenitors

M. Renzo,[1,2] O. Gottlieb,[3,2] H. S. Chan,[4,5,2] J. A. Goldberg,[6,2] A. Grichener,[1] K. Sen,[1] N. Shah,[1] E. Farag,[7] and Matteo Cantiello[2,8]

[1] *Steward Observatory, Department of Astronomy, University of Arizona, 933 N. Cherry Ave., Tucson, AZ 85721, USA*
[2] *Center for Computational Astrophysics, Flatiron Institute, 162 5th Avenue, New York, NY 10010, USA*
[3] *Department of Physics and Kavli Institute for Astrophysics and Space Research, Massachusetts Institute of Technology, Cambridge, MA 02139, USA*
[4] *JILA, University of Colorado and National Institute of Standards and Technology, 440 UCB, Boulder, CO 80309-0440, USA*
[5] *Department of Astrophysical and Planetary Sciences, University of Colorado, 391 UCB, Boulder, CO 80309, USA*
[6] *Department of Physics and Astronomy, Michigan State University, East Lansing, MI 48824, USA*
[7] *Department of Astronomy, Yale University, New Haven, CT 06511, USA*
[8] *Department of Astrophysical Sciences, Princeton University, Princeton, NJ 08544, USA*

## ABSTRACT

The understanding of the mechanism(s) by which massive stars collapse and possibly explode is rapidly maturing. Uncertainties in the structure of the stellar core at the onset of collapse are often dominant in numerical simulations, and a limited number of progenitor grids are available. This is especially true for explosions where rotation and magnetic fields play a significant or primary role. We present a grid of 113 single-star models with initial masses $M_{\rm ZAMS} = 30 - 90\,M_\odot$ and initial rigid rotation $\omega_{\rm ZAMS} = 0.5 - 0.99\,\omega_{\rm crit}$ computed at $Z = 0.001$ with the open-source stellar evolution code MESA. We adopt a 128-isotope nuclear reaction network capable of following the weak reactions deleptonizing the core during and after silicon core burning. By construction, these models experience rotationally-induced chemically-homogeneous evolution, and reach the onset of collapse ($v_{\rm infall} \lesssim -300\,{\rm km\ s^{-1}}$) with large and structured amounts of angular momentum, possibly sufficient to form accretion disks on a proto-compact object. Therefore, these progenitor structures provide a homogeneous set of models with updated input physics and improved algorithmic accuracy to understand stellar explosions of (some types of) stripped-envelope supernovae, possibly jetted and/or broad-lined, collapsars or magnetar-powered, and/or long $\gamma$-ray bursts.

*Keywords:* Gamma-ray bursts (629) — Supernovae (1668) — Stellar rotation (1629) — Stellar evolution (1599) — Massive stars (732) — High Energy astrophysics (739)

## 1. INTRODUCTION

Since the establishment of a connection between massive-star populations and supernovae (SNe, W. Baade & F. Zwicky 1934), understanding how massive stars' end their life remains a capstone problem in astrophysics. How does the iron core collapse and, at least in some cases, how does the collapse successfully revert into an explosion? In the past century, the increased cadence, wavelength coverage, and depth of time-domain astronomy have transformed this question into a family of related problems, spanning a diversity of astronomical transients (e.g., Z. Cano et al. 2017). The interest in the origin of transients and compact objects is even more timely in the era of large time-domain surveys (e.g., Rubin/LSST, Ž. Ivezić et al. 2019) and routine detections of gravitational-wave **signals from compact-object** mergers (e.g., B. P. Abbott et al. 2016; I. Mandel & F. S. Broekgaarden 2022; A. G. Abac et al. 2025; The LIGO Scientific Collaboration et al. 2026).

Recent progress in SN theory has produced unprecedented agreement among numerical simulations of neutrino-radiative transfer of core collapse (e.g., H.-T. Janka 2012; T. Melson et al. 2015; A. Burrows et al. 2025; K. Nakamura et al. 2025; A. Mezzacappa 2026; C.-H. Chen et al. 2026). For non-rotating stellar progenitors, neutrino-driven turbulent convection, sustained by continued accretion onto the shock, can lead to successful stellar explosions (e.g., T. Kuroda et al. 2018; C. D. Ott et al. 2018; A. Burrows et al. 2025). While limits of this mechanism, including the role of rotation and magnetic fields remain debated (e.g., N. Soker 2024; O.

Gottlieb et al. 2025), a consensus on the fact that uncertainties in the stellar progenitor structure are dominant is emerging (e.g., S. E. Woosley et al. 2002; A. Chieffi & M. Limongi 2013; T. Sukhbold & S. E. Woosley 2016; R. Farmer et al. 2016; M. Renzo et al. 2017; B. Müller et al. 2017; M. Limongi & A. Chieffi 2018; A. Davis et al. 2019; M. Limongi & A. Chieffi 2020; E. Laplace et al. 2021; S. E. Woosley & A. Heger 2021; C. E. Fields 2022; S. W. Bruenn et al. 2023; F. R. N. Schneider et al. 2024; M. Limongi et al. 2024; L. Boccioli & L. Roberti 2024; M. Renzo et al. 2024; M. Limongi et al. 2025; A. Burrows et al. 2025; K. Maltsev et al. 2025; A. Griffiths et al. 2026; M. A. Myers et al. 2026).

The structure of the stellar core of a progenitor model at the onset of collapse determines the initial conditions for expensive multi-dimensional, multi-physics (and often multi-million-CPU hours) simulations (e.g., H.-T. Janka 2012; B. Müller 2016; T. Takiwaki et al. 2016; D. Vartanyan et al. 2021; O. Gottlieb et al. 2025). Specifically, the profile – of density, entropy, velocity, angular momentum, magnetic fields, etc., as a function of Lagrangian mass coordinate or radius – determines the dynamics of the infalling gas and the "over-burden" caused by the accretion of outer layers through a putative nascent outflow (e.g., S. W. Bruenn et al. 2013, 2016). Therefore, it is the structure of the collapsing progenitor that decides whether a stellar model explodes or not; separately, whether it leaves behind a neutron star (NS) or a black hole (BH), either directly or through a proto-NS phase; and ultimately the properties of the outflows, the associated transient(s), and final remnants (e.g., T. Kuroda et al. 2018; C. D. Ott et al. 2018; D. Vartanyan et al. 2019; A. Burrows et al. 2024; K. Nakamura et al. 2025; A. Burrows et al. 2025).

A current bottleneck in numerical simulations of stellar explosions is the lack of models of stellar progenitors computed *i*) sufficiently late into the collapse and *ii*) including sufficient (nuclear and weak) physics to capture the known critical ingredients for the explosion. This is especially true for fast rotating progenitors of explosions where magnetic fields and angular momentum may play a dominant role.

Specifically, condition *i*) requires computing stellar evolution models well into the sub-sonic phase of the collapse, at least until the remaining time to core bounce becomes shorter than the characteristic timescales of all relevant stellar processes. This typically means evolving a stellar model beyond the point where the iron-core infall velocity $v_{\rm infall} = \min(v)$ is lower than $\lesssim -300\,{\rm km\ s^{-1}}$ (e.g., R. Farmer et al. 2021, 2023; O. Gottlieb et al. 2024) if not $-1000\,{\rm km\ s^{-1}}$ (e.g., S. E. Woosley et al. 2002). Beyond this point, matter typically becomes partially optically thick to neutrinos, and the thermodynamic quantities exceed the bounds of tabulated equations of state in stellar evolution codes. **However, already before the beginning of the collapse, several inherently multidimensional processes may play a critical role (e.g., convection, and especially rotation and magnetic fields N. Yadav et al. 2020; C. E. Fields 2022; C.-H. Chen et al. 2026). For initial conditions to simulate these in multiple dimension, criterion *i*) may need modifications to earlier times.**

For condition *ii*), stellar progenitor models used to study the explosion physics must accurately capture the progenitor's density and angular momentum structure. Any star evolving through silicon core burning will experience copious amounts of electron captures (e.g., W. D. Arnett 1977; W. R. Hix & F.-K. Thielemann 1993, 1999; A. Grichener et al. 2025; M. A. Myers et al. 2026) which progressively de-leptonize the core (i.e., decrease the amount of free electrons). The notable exceptions to this are "electron-capture" SN progenitors which collapse with an ONeMg-rich core (however, these too require appropriate treatment of the weak reactions, including URCA processes, K. Nomoto 1984; K. Takahashi et al. 2013; S. Jones et al. 2014; S. Zha et al. 2019) and progenitors above the pair-instability gap (e.g., G. Rakavy & G. Shaviv 1967; M. Renzo et al. 2024), which collapse because of the photodisintegration instability in an oxygen-rich core (e.g., J. R. Bond 1984; O. Gottlieb et al. 2025). In both these cases, the stars reach their demise bypassing the hydrostatic silicon core-burning phase.

In all other progenitors, electron captures occur at a significant rate even during the hydrostatic silicon core burning phase, before a model is sufficiently evolved according to *i*). Even adopting a "quasi-statistical equilibrium" approximation, the weak reactions that determine $Y_e$ do not reach detailed balance, since most occur when matter is still transparent to neutrinos, and the equilibrium itself depends on $Y_e$ (e.g., W. R. Hix & F.-K. Thielemann 1996). Capturing the weak interactions in sufficient detail to characterize the stellar *structure* at the onset of collapse requires considering $N \sim 100$ isotopes (e.g., R. Farmer et al. 2016; A. Grichener et al. 2025). To decrease the computing time required to calculate a grid of stellar progenitors, a commonly adopted shortcut is to use small ($N$ ∼20-isotope) nuclear reaction networks (e.g., O. R. Pols et al. 1995; F. X. Timmes et al. 2000; A. Griffiths et al. 2025). These bypass the complex nuclear and weak physics of silicon core burning by lumping all weak interactions in a custom compound reaction (see F. X. Timmes et al. 2000; P. Marchant et al. 2019; A. Grichener et al. 2025), effectively sacrific-

ing the accuracy of the core structure that decides the explosion. Even if such a compound reaction is tuned on the central value of the electron fraction $Y_{e,c}$ from models computed with larger nuclear reaction networks (e.g., D. R. Aguilera-Dena et al. 2018; F. R. N. Schneider et al. 2021, 2024), small networks cannot predict the *profile* of the electron fraction $Y_e(m)$ in the collapsing core where the explosions are determined (e.g., M. Renzo et al. 2024).

Using small nuclear reaction networks is a useful trade-off to study, for example, electromagnetic observables of the progenitor or explosions determined in the outer surface layers (e.g., V. Morozova et al. 2015; B. Paxton et al. 2018; J. A. Goldberg et al. 2019; A. Gilkis et al. 2025) or for the application of semi-analytic explodability criteria based on existing detailed simulations (e.g., B. Müller et al. 2016; F. R. N. Schneider et al. 2024; E. Laplace et al. 2025; A. Gilkis et al. 2026). However, such networks are not sufficient for investigations of explosion driving, pre-explosion and explosive neutrino signals, nuclear yields, and the resulting compact-object properties. This applies to both non-rotating (R. Farmer et al. 2016) and rotating (M. Renzo et al. 2024) progenitors. The reason is that the effective Chandrasekhar mass of the collapsing core scales with $\propto Y_e^2$, or in other words, the contribution of electron-degeneracy-pressure to the support of the core amplifies small imprecision in the core structure to a level crucial for the collapse and explosion dynamics (e.g., H.-T. Janka 2012; A. Grichener et al. 2025).

Several grids of stellar progenitors evolved sufficiently late (condition *i*) and with *detailed* nuclear physics (condition *ii*) exist, mostly from closed-source codes (e.g., KEPLER T. A. Weaver et al. 1978; S. E. Woosley et al. 2002; T. Sukhbold & S. E. Woosley 2016; T. Sukhbold et al. 2018 – FRANEC[9] M. Limongi & A. Chieffi 2003; A. Chieffi & M. Limongi 2013; M. Limongi & A. Chieffi 2018; M. Limongi et al. 2024, 2025 – HOSHI, K. Takahashi et al. 2013). While these stellar models have been invaluable in allowing the progress made so far, challenges remain in the exploration of how the results depend on the underlying *physical* and *numerical* choices in modeling the evolution of stellar progenitors. Approximations at both levels have been shown to affect the core structure and thus the explosion outcome (e.g., R. Farmer et al. 2016; M. Renzo et al. 2017; A. Davis et al. 2019; E. Laplace et al. 2021; R. Farmer et al. 2021, 2023; K. Maltsev et al. 2025; M. A. Myers et al. 2026). The few public grids of progenitors fulfilling both conditions *i*) and *ii*) computed with the open-source and community-driven code MESA (e.g., R. Farmer et al. 2016; M. Renzo et al. 2017; E. Farag et al. 2020; E. Laplace et al. 2021; R. Farmer et al. 2021, 2023; M. A. Myers et al. 2026) remain largely unexplored in explosion simulations (see however R. Perna et al. 2018; D. Vartanyan et al. 2021; O. Gottlieb et al. 2024; H.-S. Chan et al. 2026).

Despite the importance of fast-rotating progenitors for jetted explosions, collapsars, magneto-hydrodynamic-explosions, and/or long $\gamma$-ray burst (LGRB, S. E. Woosley 1993; A. I. MacFadyen & S. E. Woosley 1999; O. Gottlieb 2025), even fewer grids of such stellar models exist (e.g., A. Heger et al. 2005; R. Hirschi et al. 2004, 2005a,b,c). To the best of our knowledge, these are limited to the KEPLER models from S. E. Woosley & A. Heger (2006), which use the adaptive nuclear reaction network from A. Heger et al. (2000), and the MESA models of D. R. Aguilera-Dena et al. (2018, 2020), which adopt a small nuclear reaction network that cannot fulfill condition *ii*). Alternatively, non-rotating stellar models endowed with ad-hoc angular momentum and magnetic field profiles that are not consistent with the progenitor evolution are often used (e.g., D. M. Siegel et al. 2022; O. Gottlieb et al. 2025).

The reason for the small number of stellar progenitor grids is the substantial computational effort needed to evolve massive stars with large nuclear-reaction networks[10], coupled with the stiffness of the equations (see A. Grichener et al. 2025), which results in large fractions of progenitor stars encountering numerical issues before reaching the desired infall velocity. These can manifest at the surface or in the core (or both), and can often be an unclear mixture of poorly-resolved physical phenomena and purely algorithmic problems. Development of numerical strategies to clarify and address these issues are ongoing (e.g., B. Paxton et al. 2013; A. S. Jermyn et al. 2023; A. Grichener et al. 2025; A. Griffiths et al. 2025).

In the meantime, we present here a publicly available grid of fast-rotating models fulfilling both criteria *i*) and *ii*) for studies of the collapse and possible explosion of fast-rotating stellar cores. In Sec. 2 we describe our physical assumptions and numerical setup. Sec. 3 presents the grid of progenitors, and in Sec. 4.1 we briefly outline the astrophysical scenarios that they could ap-

[9] which does not include hydrodynamics and thus cannot reach condition *i*), thus requiring an extra step to be used as progenitors in multi-dimensional simulations (e.g., E. O'Connor & C. D. Ott 2010; M. Limongi & A. Chieffi 2020).

[10] Neglecting sparsity, the size of the matrix to solve scales as $\sim N^2$, see Appendix B in M. Renzo et al. 2017 for the exact scaling in MESA.

proximate. We contextualize our result in terms of the input and code physics in Sec. 4.2 before concluding in Sec. 5. Appendix A lists the 128 isotopes we track in our models, Appendix B provides some key quantities for the "explodability" and surface composition at the onset of collapse. Appendix C describes the input and output files of our simulations and our processing scripts publicly available at doi.org/10.5281/zenodo.14286306.

## 2. NUMERICAL SETUP

The grid of models presented here is an extension of (and includes) the progenitor star used in O. Gottlieb et al. (2024) and constitutes the input for the "averaged" progenitor constructed in H.-S. Chan et al. (2026).

We use the MESA code (version `r24.03.1`, B. Paxton et al. 2011, 2013, 2015, 2016, 2019; A. S. Jermyn et al. 2023), to compute low-metallicity ($Z = 0.001$) and fast-rotating stellar models. Our grid spans initial masses $M_{\rm ZAMS} = 30 - 100\,M_\odot$ and initial (rigid) rotation with angular frequency $\omega_{\rm ZAMS} = 0.5 - 0.99\,\omega_{\rm crit}$, where $\omega_{\rm crit} = \sqrt{(1 - L/L_{\rm Edd})GM/R^3}$ is the surface critical rotation rate accounting for radiative forces, $L$ is the stellar luminosity, $L_{\rm Edd}$ the Eddington luminosity, $G$ is the gravitational constant, $M$ is the total mass and $R$ the stellar radius. We focus on fast-rotating models to minimize numerical issues in the envelopes (e.g. P. C. Joss et al. 1973; B. Paxton et al. 2013) and produce sufficiently fast rotating core-collapse progenitors for explosions studies where angular momentum and magnetic field play a role.

We use the SKYE equation of state (A. S. Jermyn et al. 2021), and include electron screening following A. I. Chugunov et al. (2007) and thermal neutrino losses following N. Itoh et al. (1996). We adopt radiative opacities from OPAL (C. A. Iglesias & F. J. Rogers 1993, 1996) and electron conduction opacities from S. Cassisi et al. (2007).

Nuclear reaction rates are a combination of NACRE (C. Angulo et al. 1999), JINA REACLIB (R. H. Cyburt et al. 2010), and $^{12}{\rm C}(\alpha,\gamma)^{16}{\rm O}$ from R. Kunz et al. (2002), plus additional tabulated weak reaction rates G. M. Fuller et al. (1985); T. Oda et al. (1994); K. Langanke & G. Martínez-Pinedo (2000). To capture with sufficient accuracy the deleptonization of the core from the structural point of view (condition *ii*) in Sec. 1), we adopt the fully-coupled 128-isotope nuclear reaction network `mesa_128.net` (see Appendix. A and R. Farmer et al. 2016). For central temperatures above $\log_{10}(T) \geq 9.2$, we employ MESA in operator split mode, that is we iteratively solve for the structure at fixed composition, and vice versa **(e.g., Sec. 10 in A. S. Jermyn et al. 2023, see also R. Farmer et al. 2016; M. Limongi et al. 2024, 2025).**

Throughout the evolution, we include the wind mass loss rate following J. Vink et al. (2000) unless the surface $^4$He mass fraction $Y_{\rm surf} \leq 0.4$, in which case we adopt the rate from T. Nugis & H. J. G. L. M. Lamers (2000). In both cases we assume the metallicity scaling from J. S. Vink et al. (2001) and the rotational enhancement from D. B. Friend & D. C. Abbott (1986) (see also N. Langer 1998).

We use the Schwarzschild criterion and the L. Henyey et al. (1965) formulation of mixing-length theory for convective layers, with $\alpha_{\rm MLT} = 1.5$. **E. H. Anders et al. (2022) showed that for times long compared to the convective turnover timescale, convective penetration leads to the growth of the core to the Schwarzschild-defined boundary even starting from the P. Ledoux (1947)-boundary.** For convective boundary mixing, we include core overshooting as a step-function following I. Brott et al. (2011) ($\alpha_{\rm ov} = 0.335$) and treat rotational mixing in the diffusion approximation following A. Heger et al. (2000), including angular momentum transport by convection or by a "classical" Tayler-Spruit dynamo (H. C. Spruit 2002) in radiative regions.

All the models presented here result by construction in a rotationally-driven chemically homogeneous evolution (CHE, e.g., A. Maeder & G. Meynet 2000; S.-C. Yoon et al. 2006). This limits (but does not eliminate) the number of numerical issues occurring in the envelope. After the formation of a carbon-oxygen core, we prevent the development of spurious numerical velocities in the outer layers. We artificially set to zero the velocity in any layer with sound-propagation time to the surface longer than the current timestep, either from the outer edge of this core, or the location where the specific entropy drops below $s \leq 10.5\,k_B N_A$, whichever **occurs at smaller radius**. Similar artificial damping exists in most calculations of post-core-carbon burning massive stars (e.g., D. R. Aguilera-Dena et al. 2018).

To produce progenitors sufficiently close to core-bounce (condition *i*) in Sec. 1), we evolve each model until the infall velocity in the iron core decreases below $v_{\rm infall} \leq -300\,{\rm km\ s^{-1}}$ (e.g., R. Farmer et al. 2023; O. Gottlieb et al. 2024; M. A. Myers et al. 2026). This threshold is motivated by the numerical experiments performed in O. Gottlieb et al. (2024), where we showed that the physical time remaining between this point and the canonical threshold of $-1000\,{\rm km\ s^{-1}}$ is much shorter than the timescale governing the local evolution of the rotation and magnetic fields.

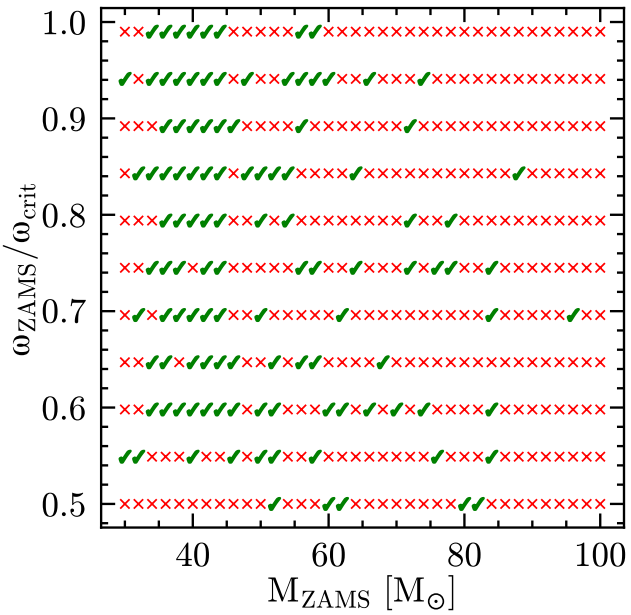



**Figure 1.** Overview of the grid success rate. Green checkmarks correspond to successful models. Red crosses correspond to models that either do not reach infall velocities of $-300\,\mathrm{km\ s^{-1}}$ or they exhibit large luminosity and effective temperature excursions that are treated unphysically in our setup and can impact the core-structure. **Other colors indicate the initial rotation rate following the color bar in Fig. 5.**

Spatial and temporal resolution should always be tested in numerical models of stellar evolution (e.g., R. Farmer et al. 2016). We refer readers to the appendix of O. Gottlieb et al. (2024) for resolution tests performed with the same setup and code on the initially $40\,M_\odot$ and $\omega = 0.6\,\omega_{\rm crit}$ model.

## 3. RESULTS

Our grid consists initially of 11 values of $\omega_{\rm ZAMS}/\omega_{\rm crit}$ times 71 $M_{\rm ZAMS}$ values, for a total of 781 stellar models (Fig. 1). Of these, 113 successfully reach the onset of core collapse and pass our visual inspection to remove models experiencing large $L$ and effective temperature ($T_{\rm eff}$) excursions. These may be manifestations of physical processes in the envelope (e.g., A. Heger et al. 1997; J. Fuller 2017 for red supergiants and e.g, J. Fuller & S. Ro 2018 for envelope-less stars) and/or numerical artifacts amplified by the small timesteps necessary in late evolutionary phases (since the acceleration term in the momentum equation is calculated as $\Delta v/\Delta t$, it can become extremely large for small timesteps $\Delta t$, B. Paxton et al. 2011).

Because of the fast initial rotation and assumed rotational mixing, our models do not develop extended hydrogen envelopes, and our setup is not designed to properly capture physical phenomena in the outer layers this late in the evolution. Furthermore, large $L$ and $T_{\rm eff}$ excursions feed back nonlinearly on the core in structurally significant ways, motivating our choice to exclude models exhibiting this numerical behavior from our grid. Overall, our grid has a 14% success rate, relatively high for models that are *not* manually mentored in any way. **Below, we focus on describing the overall properties of our grid. We refer interested readers to M. Renzo et al. (2024) and the appendix in O. Gottlieb et al. (2024) for a more detailed description of one example model.**

As shown in Fig. 1, successful models in our grid (green checkmarks) sample relatively well, the parameter space $30\,M_\odot \lesssim M_{\rm ZAMS} \lesssim 50\,M_\odot$ across all rotation rates explored ($0.5 \lesssim \omega_{\rm ZAMS}/\omega_{\rm crit} \lesssim 0.99$), and the number of successes decreases at larger initial masses and rotation rates. However, there is no particular threshold beyond which models systematically fail: this suggests most issues are related to numerical errors in the solution of the stiff equations rather than physical hypothesis.

From this point onward, we focus on the analysis of the successful models. Fig. 2 shows the evolution on the Herzsprung-Russel diagram of our rotationally-induced CHE models with each thin line corresponding to one star. Colors indicate the initial $\omega_{\rm ZAMS}/\omega_{\rm crit}$, with darker colors corresponding to values close to 0.5 and lighter colors for faster initial rotation (see also color bar in Fig. 5). The thick orange lines correspond to the $40\,M_\odot$, $\omega_{\rm ZAMS} = 0.6\,\omega_{\rm crit}$ model used as progenitor in O. Gottlieb et al. (2024), highlighted as a guideline. The thin black dot-dashed line is the same model computed with a 22-isotope network from M. Renzo et al. (2024) available from doi.org/10.5281/zenodo.11375522.

The stars evolve from the bottom right toward the top left. Because of the large rotation imposed at birth, these models experience CHE (A. Maeder & G. Meynet 2000; S.-C. Yoon et al. 2006; S. E. de Mink et al. 2009). Rotational mixing (dominated by meridional circulations) prevents the formation of a chemical stratification and the emergence of a core–envelope boundary during the hydrogen-core burning main sequence. The entire star is enriched in $^4$He (and $^{14}$N at the expense of $^{12}$C and $^{16}$O because of the CNO cycle, H. A. Bethe 1939), reducing the envelope opacity and preventing the redward evolution and radial expansion. Nevertheless, the increase in mean molecular weight **over time** leads to an increase in luminosity (e.g., R. Kippenhahn et al. 2013; C. Xin et al. 2022).

At the end of the hydrogen-core burning main sequence, each track shows a characteristic "hook": this

corresponds to a thermal timescale phase of contraction because of the exhaustion of hydrogen in the entire core region, corresponding to most of the star in these CHE models.

Helium ignition in the center causes the star to briefly re-expand up to its maximum radius. Nevertheless, each star remains relatively small, with $\max(R) \lesssim 31\,R_\odot$ and $T_{\rm eff} \gtrsim 10^{4.5}\,{\rm K}$ across the entire grid. This minimizes the chances for binary mass-transfer with putative companions for CHE stars (e.g., S. E. de Mink et al. 2009; S. E. de Mink & I. Mandel 2016; I. Mandel & S. E. de Mink 2016; P. Marchant et al. 2016). We emphasize that from this point onwards, the evolution is not *homogeneous* anymore: while rotational mixing remains active, the evolutionary timescale speeds up significantly owing to the lower energy release per nucleon for heavier nuclear fuels, and chemical gradients seeded by the burning impede further mixing (e.g., A. Heger et al. 2000; S.-C. Yoon et al. 2006; A. Chieffi & M. Limongi 2013). The stars develop a distinct carbon-oxygen core, and an "onion"-like structure, but lack a Hydrogen-rich, expanded envelope.

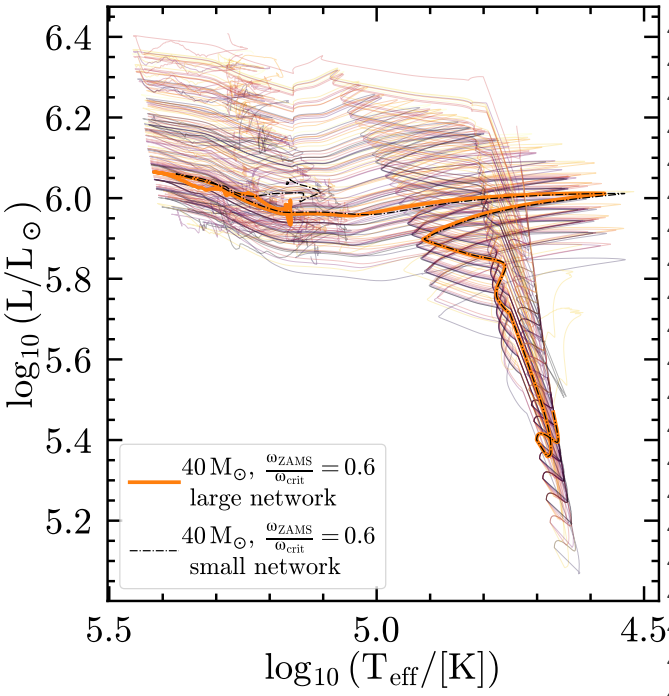



**Figure 2.** Herzsprung-Russel diagram of rotationally-induced chemically-homogeneous evolving single star models. Dark (light) colors correspond to lower (higher) initial rotation rates. The orange thick line marks a representative $40 M_\odot$, $\omega_{\rm ZAMS} = 0.6\,\omega_{\rm crit}$ used in O. Gottlieb et al. (2024), and the thin black dot-dashed line shows an otherwise identical model computed with only 22-isotopes from M. Renzo et al. (2024). **Other colors indicate the initial rotation rate following the color bar in Fig. 5.**

The remaining evolution occurs roughly right-to-left at constant luminosity, with wind mass-loss progressively revealing hotter inner layers of the star. The hottest point and smallest radius is reached roughly at core Neon ignition (O. Gottlieb et al. 2024), after which stars typically expand slightly again because of the enhanced core contraction as neutrino cooling and nuclear neutrino losses increase. We note that the majority of the **unsuccessful** models (red crosses in Fig. 1, excluded from Fig. 3 and beyond) **fail** during or after Neon core burning. This is because this burning phase occurs through photodisintegration $^{20}{\rm Ne}(\gamma,\alpha)^{16}{\rm O}$ which produces $\alpha$ particles. These have extremely high reaction rates at the relevant temperatures $T \gtrsim 1.5\times10^9\,{\rm K}$, (R. Kippenhahn et al. 2013), thus the equations for the evolution of the core composition become significantly stiffer and numerically challenging.

The comparison between the two $40\,M_\odot$ and $\omega_{\rm ZAMS} = 0.6\,\omega_{\rm crit}$ models computed with a 128-isotope (thick orange solid) and 22-isotope (thin black dot-dashed) nuclear reaction network shows how the surface $L$ and $T_{\rm eff}$ are relatively insensitive to the treatment of very late burning. Minor differences occur only during thermal-timescale phases (e.g., coldest point during the "hook") and at the very end, when the outer layers of these envelope-less stars can be dynamically and/or numerically coupled to the core.

Fig. 3 shows an overview of their evolution and structure at the onset of core collapse. The top left panel shows the specific entropy profiles $s$ of the innermost $m \leq 4\,M_\odot$. Nearly flat "steps" correspond to stellar regions that have experienced *efficient* convection less than a local thermal time-scale before the onset of collapse. This results in a very nearly adiabatic (i.e., constant $s$) profile without enough time left to relax. The mass location of these depends on the detailed evolution of each stellar model in our grid. The vertical axis is cut at $s = 15\,k_B N_A$ to focus on visualizing the inner core where stellar explosions are decided, and thus excludes the outer (higher-entropy) layers of the stars. The comparison between the solid orange line and the thin dash-dotted black line for the two $40\,M_\odot$, $\omega_{\rm ZAMS} = 0.6\,\omega_{\rm crit}$ models differing only in size of the nuclear network shows very significant differences in the layers where explosions are decided. In this (and other) panels, the differences introduced by the simplification of the weak interactions is comparable or larger than the difference between pairs of models of different mass and initial rotation computed with large nuclear reaction networks. Thus, the systematic error introduced by a small nuclear reaction network

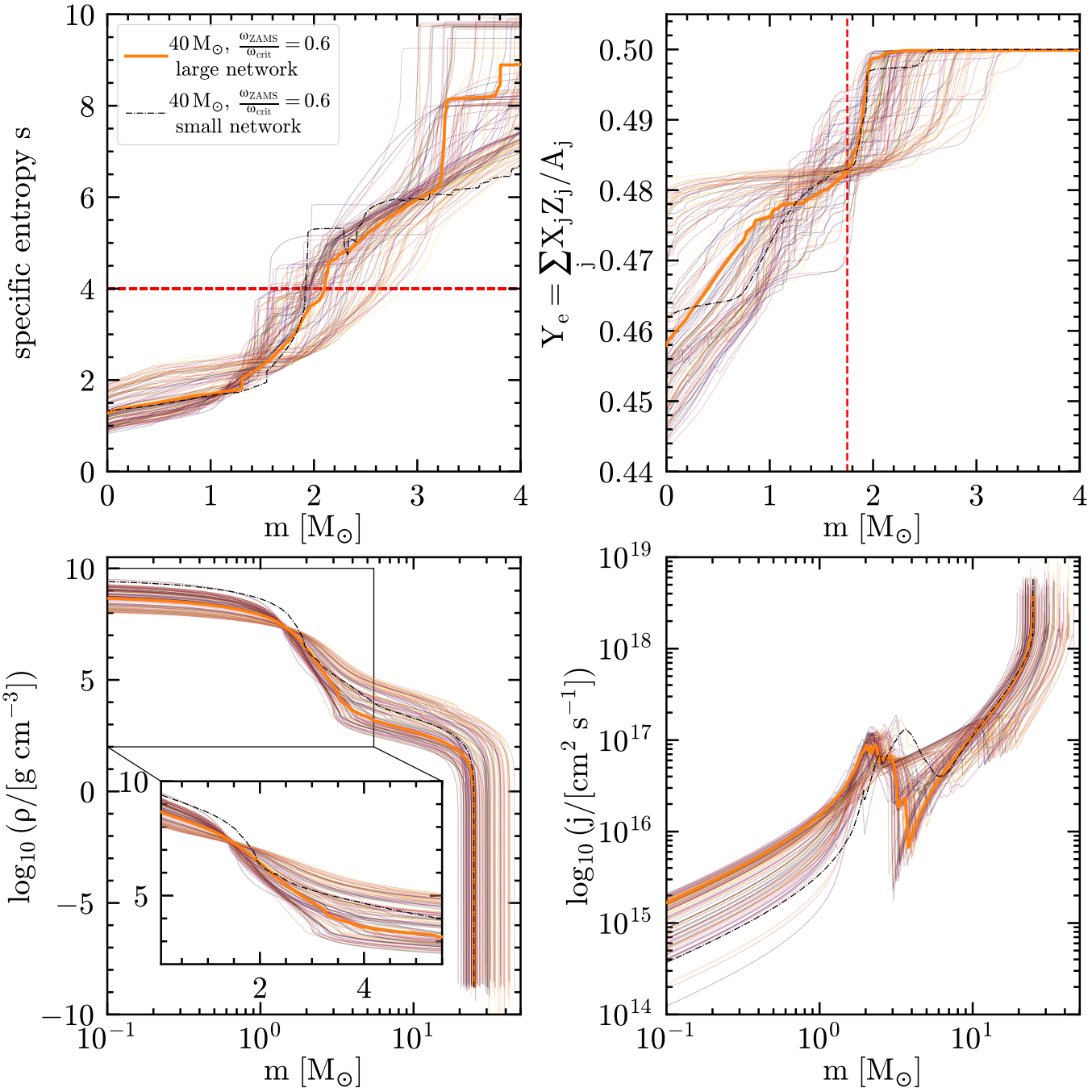



**Figure 3.** *Top left*: Inner specific entropy profiles. The red dashed horizontal line marks the $s = 4\,k_B N_A$ location often used in "explodability" criteria (e.g., T. Ertl et al. 2016; T. Ertl et al. 2020). *Top right:* Inner electron fraction $Y_e(m)$ profile as a function of mass. The red dashed vertical line marks $\mathcal{M} = 1.75\,M_\odot$ used to calculate the compactness $\xi_{1.75}$. *Bottom left:* Density profiles as a function of Lagrangian mass coordinate $m$ on a logarithmic scale. The inset zooms on the inner region and uses a linear scale. *Bottom right:* Specific angular momentum profiles as a function of $m$. In all panels, dark (light) colors correspond to lower (higher) initial rotation rates. the orange thick line marks a representative $40 M_\odot$, $\omega_{\rm ZAMS} = 0.6\,\omega_{\rm crit}$ used in O. Gottlieb et al. (2024), and the thin black dot-dashed line shows an otherwise identical model computed with only 22-isotopes from M. Renzo et al. (2024). **Other colors indicate the initial rotation rate following the color bar in Fig. 5.**

biases inferences about the connection between progenitors, explosions, and remnants (R. Farmer et al. 2016; M. Renzo et al. 2024).

The top right panel of Fig. 3 shows the profiles of the electron fraction $Y_e = \sum_j X_j Z_j / A_j$, where $X_j$ is the mass fraction of the $j$-th isotope with charge $+Z_j e$ and atomic mass $A_j$, in the innermost mass coordinate $m \leq 4\,M_\odot$ at the onset of core collapse. Outside this mass coordinate $Y_e \simeq 0.5$, while inside, weak reactions deleptonize the core and lower the $Y_e$, creating a profile consistent with the thermodynamics in the evolving cores.

The bottom left panel of Fig. 3 depicts the full density profiles at the onset of collapse as a function of $\log_{10}(m/M_\odot)$, to focus on the inner layers of our CHE models. The inset panel shows the innermost mass coordinate $m < 5\,M_\odot$ on a linear scale. The density profile $\rho(m)$ in these inner layers depends strongly on the $Y_e(m)$ because of the contribution of electron degeneracy pressure to the support of the core, and it determines the infall velocity profile once electron captures (and later photodisintegration) remove pressure support (e.g., H.-T. Janka 2012). Moreover, the density determines the initial accretion onto the shock generated at core bounce **and thus the ram pressure impinging on the shock** and the neutrino luminosity (e.g. D. Vartanyan et al. 2021; A. Burrows et al. 2024). Finally, the density profile is also related to the angular momentum profile via the mass-continuity equation relating $\rho(m)$ and the radial distribution of mass $r(m)$. The comparison between the two $40\,M_\odot$, $\omega = 0.6\,\omega_{\rm crit}$ models suggests that the collapse dynamics can be completely different in these, creating a large systematic uncertainty on explosion models when relying on small nuclear reaction networks (e.g., M. Renzo et al. 2024).

The bottom right panel of Fig. 3 shows the specific angular momentum per unit mass $j = r^2\omega$ at the onset of core collapse. This can be compared to the angular momentum of the innermost stable circular orbit of a putative seed BH to decide whether a disk forms before (e.g., O. Gottlieb et al. 2024) or during a proto-NS phase (e.g., D. Kasen & L. Bildsten 2010), or after the formation of the BH (e.g., A. I. MacFadyen & S. E. Woosley 1999; R. Perna et al. 2018). With our modeling assumptions, disk formation before BH formation appears to be more common in our grid, meaning our models retain a large amount of angular momentum at the onset of collapse.

The Tayler-Spruit dynamo responsible for angular momentum transport in radiative regions of our CHE models generates magnetic fields. Fig. 4 shows the azimuthal ($B_\varphi$, top) and the radial ($B_r$, bottom) components as a function of mass coordinate on a logarithmic scale at the onset of collapse. The latter are *local* and not large-scale coherently organized fields. We refer interested readers to O. Gottlieb et al. (2024) (Sec. 2.1) for more details and how to calculate the magnetic flux advected during the initial phases of a core collapse.

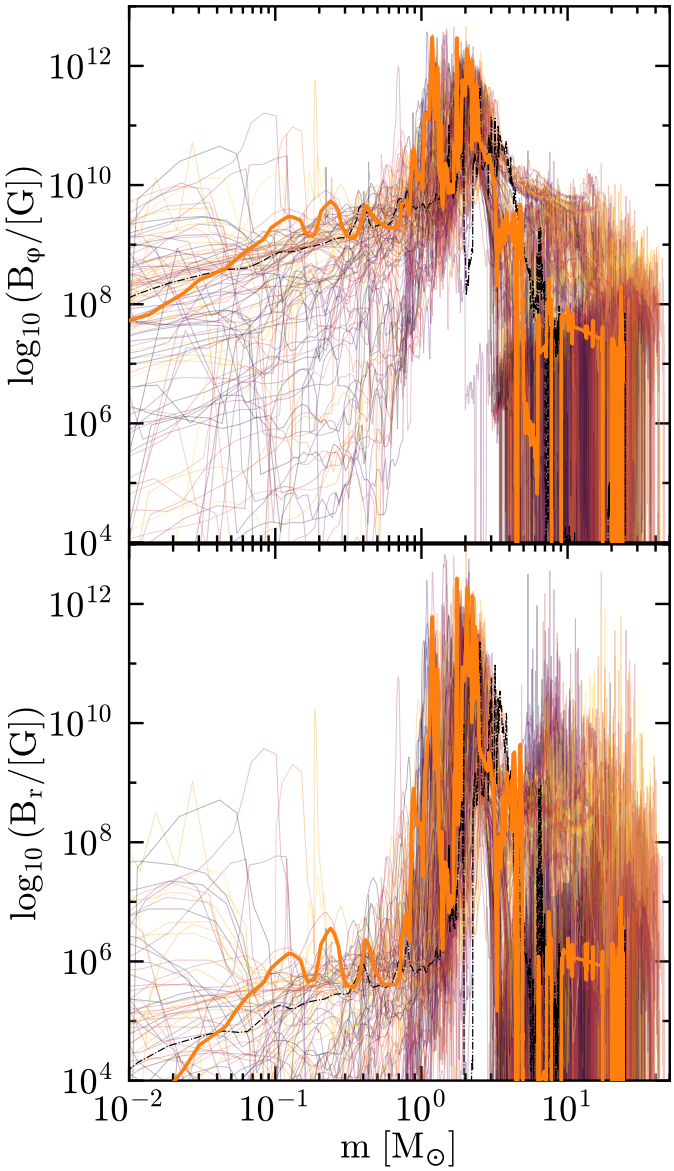



**Figure 4.** Azimuthal (top) and radial (bottom, local) component of the magnetic field generated by the Tayler-Spruit dynamo as a function of Lagrangian mass coordinate. Dark (light) colors correspond to lower (higher) initial rotation rates. The orange thick line shows the $40 M_\odot$, $\omega_{\rm ZAMS} = 0.6\,\omega_{\rm crit}$ model from O. Gottlieb et al. (2024) and the thin black dot-dashed line shows an otherwise identical model computed with only 22-isotopes from M. Renzo et al. (2024). **Other colors indicate the initial rotation rate following the color bar in Fig. 5.**

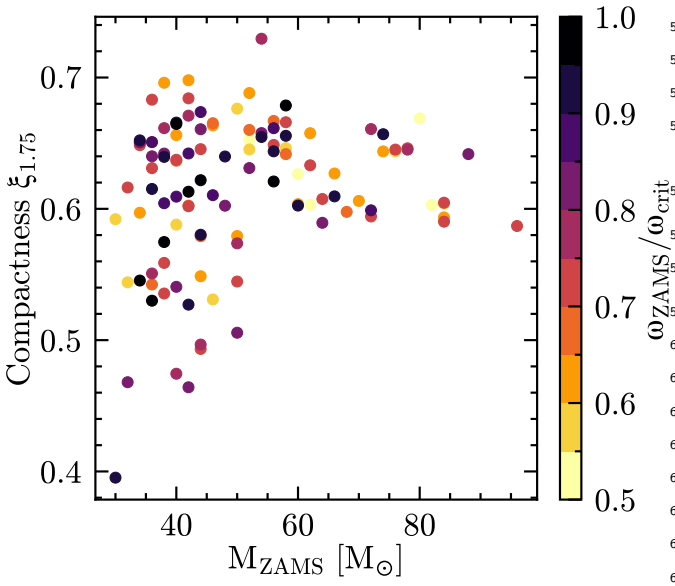



**Figure 5.** Compactness as a function of initial mass and rotation rate for single CHE models. Higher values of compactness $\xi_{\mathcal{M}}$ favor BH formation, but this does not imply without outflows and/or electromagnetic transients (e.g., A. Burrows et al. 2025; H.-S. Chan et al. 2026). **Colors indicate the initial rotation rate.**

Fig. 5 shows the compactness parameter (E. O'Connor & C. D. Ott 2011):

$$\xi_{\mathcal{M}} = \frac{\mathcal{M}/M_\odot}{r(\mathcal{M})/1000\,\mathrm{km}} \quad , \tag{1}$$

against the initial total mass $M_{\rm ZAMS}$. We use $\mathcal{M} = 1.75\,M_\odot$ to calculate the compactness following A. Burrows et al. (2024) (see also M. Ugliano et al. 2012), highlighted also by the vertical black dashed line in the top right panel of Fig. 3. Large values of $\xi_{1.75}$ are more likely to result in BH formation, but this does not imply that the collapse does not result in a successful explosion since the accretion rate on the shock, and consequently the neutrino luminosity, thought to drive the shock, correlates positively with $\xi_{1.75}$ **in multi-dimensional simulations** (e.g., T. Kuroda et al. 2018; C. D. Ott et al. 2018; A. Burrows et al. 2025). Overall, slower rotation (darker colors) tends to result in higher compactness, while increasing rotational support decreases the core density slightly and decreases $\xi_{1.75}$, but the trend is not always respected, likely due to the high non-linearity of stellar evolution.

We report in Tab. 1 the values of the mass coordinate where the specific entropy drops below $s = 4\,k_B N_A$ ($M_4$) and the mass gradient at that location $\mu_4 = dm/dr(m = M_4)$ which can be used to construct a two-parameter explodability criterion (T. Ertl et al. 2016; T. Ertl et al. 2020). Unsurprisingly for the mass regime considered here, all models lie above the critical line for BH formation, regardless of the adopted engine calibration. Nevertheless, some of our models may retain too much angular momentum to directly collapse into a BH.

## 4. DISCUSSION

### 4.1. *What astrophysical scenario can these models represent?*

Direct observational evidence (e.g., from stellar spectroscopy) for rotationally-induced CHE is still debated (e.g., I. Hunter et al. 2008; F. Martins et al. 2009, 2013; L. A. Almeida et al. 2015; L. Mahy et al. 2020; K. Sharpe et al. 2024), but this evolutionary path becomes more likely at very high stellar masses and low metallicities (e.g., S.-C. Yoon et al. 2006), for which samples are sparser (e.g., C. J. Evans et al. 2011; T. Shenar et al. 2024).

To produce fast-rotating collapsar progenitors, we construct our stellar models with initial metallicity $Z = 0.001$ and *assuming* extremely fast initial rotation rates. Such initial rotations are prohibitively rare regardless of $Z$ in young massive star populations in the local Universe (e.g., I. Hunter et al. 2008; O. H. Ramírez-Agudelo et al. 2013; O. H. Ramírez-Agudelo et al. 2015; P. L. Dufton et al. 2019; D. Galán-Diéguez et al. 2026; D. J. Lennon et al. 2026). While there is growing evidence for relatively fast ($\gtrsim 250\,\mathrm{km\ s^1}$) primordial rotation (e.g., Y. Nazé et al. 2023; N. Britavskiy et al. 2024; Y. Nazé et al. 2025), these still have $\omega \lesssim 0.3 - 0.4\,\omega_{\rm crit}$ and are not as fast as we assume here. The fastest-rotating known O-type stars reach $v\sin(i) \sim 600\,\mathrm{km\ s^{-1}}$, likely in the regime considered here, and are most likely accretor stars in interacting binaries (M. Cantiello et al. 2007; O. H. Ramírez-Agudelo et al. 2015; M. Renzo & Y. Götberg 2021). The accretion process can change the core-envelope boundary in ways that are not accounted for in our single-star models (e.g., M. Cantiello et al. 2007; M. Renzo & Y. Götberg 2021; M. Renzo et al. 2023; T. Wagg et al. 2024; C. Landri et al. 2025; K. Nathaniel et al. 2025; N. Shah et al. 2026; J. Henneco & D. M. Bowman 2026).

Nevertheless, observations of jetted explosions and LGRBs suggest that at least some stars reach core collapse with sufficient angular momentum to form accretion disks onto a seed compact object. This could be the product of accretion on the secondary in an interacting binary system (e.g., M. Cantiello et al. 2007), stellar mergers retaining angular momentum (e.g., E. Chatzopoulos et al. 2020; M. Renzo et al. 2020), tidal spin up in post-common envelope massive binaries (S. S. Bavera et al. 2020, 2022; K. Sen et al. 2025), or possibly inefficient angular momentum transport in stars, leading

to spin up of the core as it evolves and contracts. We do not consider here the possibility of disk formation from stochastic angular momentum in extended convective envelopes (e.g., E. Quataert et al. 2019; D. Shishkin & N. Soker 2021; A. Antoni et al. 2025) because our progenitors have small radii. LGRBs are associated with type Ic broad-line SNe that require hydrogen-free progenitors with compact envelopes. We note however that all our progenitors retain helium at their surfaces (cf. Tab. 1), a result certainly influenced by the assumed wind mass loss (e.g., M. Renzo et al. 2017).

We do not attempt to investigate the detailed evolutionary scenarios leading to fast-rotating cores. Instead, we provide a computationally homogeneous grid of single, initially fast-rotating stellar progenitors that could *approximate* these scenarios. The grid is designed to enable systematic numerical experiments on stellar collapse in the regimes where rotation and magnetic field are expected to play an important role.

### 4.2. *Known physical and numerical caveats*

The **rotationally-induced CHE path (e.g., A. Maeder & G. Meynet 2000; S.-C. Yoon et al. 2006) and the resulting profiles at the onset of collapse are** a product of the assumed initial rotation and the interplay between wind-mass loss and angular momentum transport **within the limited framework of 1D treatment of these processes (e.g., J. S. G. Mombarg et al. 2024; M. Rieutord et al. 2025).** Both stellar winds and angular momentum transport processes remain highly uncertain in stellar context (see, e.g., J. Puls et al. 2008; N. Smith 2014; J. S. Vink 2015; M. Renzo et al. 2017; J. Josiek et al. 2024; A. Romagnolo et al. 2026 for wind-related uncertainties, H. C. Spruit 1999; M. Cantiello et al. 2014; J. Fuller et al. 2019; J. W. den Hartogh et al. 2020; V. A. Skoutnev & A. M. Beloborodov 2024, 2025 for angular momentum transport uncertainties). **Other stellar physics ingredients influencing the size of the convective main-sequence core (e.g., overshooting), can also have an secondary role by changing the amount of mass that needs to be mixed to reach CHE.**

**In our spherically-symmetric models, we treat rotation in the "shellular approximation" (e.g., A. Maeder & G. Meynet 2000), which may not accurately represent the multi-dimensional flows at rotation rates** $\omega \gtrsim 0.8\,\omega_{\rm crit}$**.** We adopt a Tayler-Spruit dynamo in radiative regions (H. C. Spruit 2002) for angular momentum transport. As noted in O. Gottlieb et al. (2024) based on the $40\,M_\odot$ model with $\omega_{\rm ZAMS} = 0.6\,\omega_{\rm crit}$ (orange line in Fig. 3-4), the stronger magnetic field saturation of the Tayler-Spruit instability suggested in J. Fuller et al. (2019); J. Fuller & W. Lu (2022) still results in CHE, but at the onset of core collapse, the cores are too slowly rotating to produce accretion disks. **Lowering the initial rotation rate (see Sec. 4.1) and/or the efficiency of mixing (of composition and angular momentum) and/or increasing the early angular momentum losses could prevent rotationally-induced CHE and lead to a redward expansion and a clear core-envelope structure.**

For line-driven wind mass loss, we assumed a combination of J. Vink et al. (2000); J. S. Vink et al. (2001) and T. Nugis & H. J. G. L. M. Lamers (2000), as opposed to most previous grids of fast-rotating progenitors adopting a combination of H. Nieuwenhuijzen & C. de Jager (1990) (e.g., S. E. Woosley & A. Heger 2006) and W.-R. Hamann et al. (1995) (e.g., D. R. Aguilera-Dena et al. 2018, 2020). **The lack of extended low-density hydrogen-rich envelopes removes the need for a "cool" wind mass loss, and also allows us to avoid known numerically challenging issues (e.g., B. Paxton et al. 2013; S. J. Cheng et al. 2024).**

The 128-isotope nuclear reaction network we adopt (see also Appendix A) is sufficient to resolve the density structure of the core (e.g., R. Farmer et al. 2016). Increasing the network size further has a negligible impact on the electron fraction $Y_e$. This has been shown both with full stellar model calculations (e.g., R. Farmer et al. 2016) and focused one-zone experiments (e.g., A. Grichener et al. 2025). However, the list of isotopes adopted here may still under-resolve the pre-collapse high-energy neutrino flux (e.g., K. M. Patton et al. 2017; E. Farag et al. 2020; M. A. Myers et al. 2026) and **may be missing important seed nuclei for the subsequent explosive** nucleosynthesis (e.g., K. Nomoto & M. Hashimoto 1986; W. R. Hix & F.-K. Thielemann 1996; F.-K. Thielemann et al. 1996; S. E. Woosley et al. 2002; T. Yoshida et al. 2005; S. E. Woosley 2007; R. Farmer et al. 2023; M. Limongi et al. 2025).

Moreover, the nuclear reaction rates adopted (specifically the $^{12}{\rm C}(\alpha,\gamma)^{16}{\rm O}$ from R. Kunz et al. 2002) are under active scrutiny (e.g., R. J. deBoer et al. 2017; K. Takahashi 2018; R. Farmer et al. 2019; G. Costa et al. 2021; S. E. Woosley & A. Heger 2021; Y. Shen et al. 2023; A. Noll et al. 2025; H. Tong et al. 2026). Our choice of rates therefore inevitably influences the pre-collapse structures presented here, as is true for any stellar models in the literature (see also C. E. Fields et al. 2018). Finally, we also employ the "split-burn" mode of MESA, that is we solve separately for the composition (at fixed structure), and iterate on the structure

(at fixed updated composition). While this approach is common in stellar evolution because of the large difference in the timescale for nuclear burning and other processes, it may introduce small numerical artifacts (e.g., K. Takahashi et al. 2013; R. Farmer et al. 2016; A. S. Jermyn et al. 2023; M. Limongi et al. 2024, 2025).

As with any set of stellar evolution calculations, the representation of the inherently three-dimensional convective flow (e.g., A. S. Jermyn et al. 2022; M. Joyce & J. Tayar 2023) and convective boundaries (e.g., A. Davis et al. 2019; M. A. Myers et al. 2026) are potential causes of concern. The key hypothesis of subsonic flows holds throughout the evolution of our models. However, our choice of free parameters ($\alpha_{\rm MLT}$ and $\alpha_{\rm ov}$) to match existing literature (e.g., I. Brott et al. 2011; D. R. Aguilera-Dena et al. 2018, 2020) may not be appropriate in every stellar environment (e.g. M. Joyce & B. Chaboyer 2018; S.-H. Chun et al. 2018) or for every convective layer in a stars (e.g., J. A. Goldberg et al. 2022; J.-Z. Ma et al. 2025b; M. Stuck et al. 2025; A. Griffiths et al. 2026), and we do not include the latest advances from stellar hydrodynamic simulations (e.g. N. Yadav et al. 2020; W. C. Schultz et al. 2020; E. H. Anders et al. 2022; F. Rizzuti et al. 2023; C. Johnston et al. 2024). Perhaps more importantly, our models adopt an artificial strategy to expunge envelope velocities in the outer layers which are possibly seeded by numerical noise in the fast-paced late burning evolution. Significant efforts to improve MESA are aimed at addressing this issue.

## 5. CONCLUSION

We presented a grid of rotationally-driven chemically-homogeneously evolving massive stars at the onset of core collapse. We have described their evolution on the Herzsprung-Russel diagram and characterized their internal profiles at collapse, and discussed physical and numerical limitations of our calculations.

The evolution of these initially fast-rotating models represents a presumably extremely rare evolutionary path based on observations of initial rotation rates in the local Universe and rates of astrophysical transients. Our models could approximate, to a certain degree, more common evolutionary paths where rotation is produced by binary interactions. Since binary interactions have been shown to introduce significant differences in progenitor structures compared to single stellar evolution (e.g., E. Laplace et al. 2021; D. Vartanyan et al. 2021; R. Farmer et al. 2023; E. Laplace et al. 2025; K. Maltsev et al. 2025; J.-Z. Ma et al. 2025a), further studies to compare the outcome of these pathways with single-fast rotating models are necessary.

Despite the inevitable algorithmic limitations inherent in numerical simulations that depend on input physics and processes under active investigation (especially when represented by very stiff sets of equations), our progenitor grid constitutes a significant advance in both input physics and numerical accuracy for rapidly-rotating stars are the onset of core collapse. By providing a homogeneous set of pre-collapse models with substantial angular momentum, this grid offers useful initial conditions for numerical studies of rotating hydrogen-free massive-star explosions, including magnetohydrodynamic-driven and jetted explosions such as collapsars and long $\gamma$-ray bursts. We hope that our publicly available results (at doi.org/10.5281/zenodo.14286306) will be adopted in future numerical studies of the explosions of rotating massive stars.

*Software:* This research has made use of the Astrophysics Data System, funded by NASA under Cooperative Agreement 80NSSC21M00561. We used the following software packages: MESA (B. Paxton et al. 2011, 2013, 2015, 2019; A. S. Jermyn et al. 2023), `python` (G. Van Rossum & F. L. Drake 2009), `numpy` (C. R. Harris et al. 2020), `scipy` (P. Virtanen et al. 2020; R. Gommers et al. 2023), and `matplotlib` (J. D. Hunter 2007). Software citation information aggregated using `The Software Citation Station` (T. Wagg & F. S. Broekgaarden 2024; T. Wagg et al. 2025).

## ACKNOWLEDGEMENTS

MR is grateful to N. Britavskyi for many inspiring conversations on stellar rotation, and was partially supported by NSF-AST-2510584. J.A.G. acknowledges financial support from NASA grant 23-ATP23-0070. **We thank the referee for helpful feedback.**

## REFERENCES

Abac, A. G., Abouelfettouh, I., Acernese, F., et al. 2025, ApJL, 995, L18, doi: 10.3847/2041-8213/ae0c06

Abbott, B. P., Abbott, R., Abbott, T. D., et al. 2016, Phys. Rev. Lett., 116, 061102, doi: 10.1103/PhysRevLett.116.061102

Aguilera-Dena, D. R., Langer, N., Antoniadis, J., & Müller, B. 2020, http://ascl.net/2008.09132

Aguilera-Dena, D. R., Langer, N., Moriya, T. J., & Schootemeijer, A. 2018, ApJ, 858, 115, doi: 10.3847/1538-4357/aabfc1

Almeida, L. A., Sana, H., de Mink, S. E., et al. 2015, ApJ, 812, 102, doi: 10.1088/0004-637X/812/2/102

Anders, E. H., Jermyn, A. S., Lecoanet, D., & Brown, B. P. 2022, ApJ, 926, 169, doi: 10.3847/1538-4357/ac408d

Angulo, C., Arnould, M., Rayet, M., et al. 1999, NuPhA, 656, 3, doi: 10.1016/S0375-9474(99)00030-5

Antoni, A., Jiang, Y.-F., & Quataert, E. 2025, arXiv e-prints, arXiv:2509.16308, doi: 10.48550/arXiv.2509.16308

Arnett, W. D. 1977, ApJS, 35, 145, doi: 10.1086/190472

Baade, W., & Zwicky, F. 1934, Proceedings of the National Academy of Science, 20, 254, doi: 10.1073/pnas.20.5.254

Bavera, S. S., Fragos, T., Qin, Y., et al. 2020, A&A, 635, A97, doi: 10.1051/0004-6361/201936204

Bavera, S. S., Fragos, T., Zapartas, E., et al. 2022, A&A, 657, L8, doi: 10.1051/0004-6361/202141979

Bethe, H. A. 1939, Physical Review, 55, 434, doi: 10.1103/PhysRev.55.434

Boccioli, L., & Roberti, L. 2024, Universe, 10, 148, doi: 10.3390/universe10030148

Bond, J. R. 1984, in Astrophysics and Space Science Library, Vol. 109, Stellar Nucleosynthesis, ed. C. Chiosi & A. Renzini, 297, doi: 10.1007/978-94-009-6348-1_23

Britavskiy, N., Renzo, M., Nazé, Y., Rauw, G., & Vynatheya, P. 2024, A&A, 684, A35, doi: 10.1051/0004-6361/202348484

Brott, I., de Mink, S., Cantiello, M., et al. 2011, \aap, 530, A115, doi: 10.1051/0004-6361/201016113

Bruenn, S. W., Mezzacappa, A., Hix, W. R., et al. 2013, ApJL, 767, L6, doi: 10.1088/2041-8205/767/1/L6

Bruenn, S. W., Lentz, E. J., Hix, W. R., et al. 2016, ApJ, 818, 123, doi: 10.3847/0004-637X/818/2/123

Bruenn, S. W., Sieverding, A., Lentz, E. J., et al. 2023, ApJ, 947, 35, doi: 10.3847/1538-4357/acbb65

Burrows, A., Wang, T., & Vartanyan, D. 2024, ApJL, 964, L16, doi: 10.3847/2041-8213/ad319e

Burrows, A., Wang, T., & Vartanyan, D. 2025, ApJ, 987, 164, doi: 10.3847/1538-4357/addd04

Cano, Z., Wang, S.-Q., Dai, Z.-G., & Wu, X.-F. 2017, Advances in Astronomy, 2017, 8929054, doi: 10.1155/2017/8929054

Cantiello, M., Mankovich, C., Bildsten, L., Christensen-Dalsgaard, J., & Paxton, B. 2014, ApJ, 788, 93, doi: 10.1088/0004-637X/788/1/93

Cantiello, M., Yoon, S.-C., Langer, N., & Livio, M. 2007, A&A, 465, L29, doi: 10.1051/0004-6361:20077115

Cassisi, S., Potekhin, A. Y., Pietrinferni, A., Catelan, M., & Salaris, M. 2007, ApJ, 661, 1094, doi: 10.1086/516819

Chan, H.-S., Gottlieb, O., Jacquemin-Ide, J., Cantiello, M., & Renzo, M. 2026, arXiv e-prints, arXiv:2605.12931, doi: 10.48550/arXiv.2605.12931

Chatzopoulos, E., Frank, J., Marcello, D. C., & Clayton, G. C. 2020, ApJ, 896, 50, doi: 10.3847/1538-4357/ab91bb

Chen, C.-H., Lentz, E. J., Hix, W. R., et al. 2026, arXiv e-prints, arXiv:2604.09906, doi: 10.48550/arXiv.2604.09906

Cheng, S. J., Goldberg, J. A., Cantiello, M., et al. 2024, ApJ, 974, 270, doi: 10.3847/1538-4357/ad701e

Chieffi, A., & Limongi, M. 2013, ApJ, 764, 21, doi: 10.1088/0004-637X/764/1/21

Chugunov, A. I., Dewitt, H. E., & Yakovlev, D. G. 2007, PhRvD, 76, 025028, doi: 10.1103/PhysRevD.76.025028

Chun, S.-H., Yoon, S.-C., Jung, M.-K., Kim, D. U., & Kim, J. 2018, ApJ, 853, 79, doi: 10.3847/1538-4357/aa9a37

Costa, G., Bressan, A., Mapelli, M., et al. 2021, MNRAS, 501, 4514, doi: 10.1093/mnras/staa3916

Côté, B., Jones, S., Herwig, F., & Pignatari, M. 2020, ApJ, 892, 57, doi: 10.3847/1538-4357/ab77ac

Cyburt, R. H., Amthor, A. M., Ferguson, R., et al. 2010, ApJS, 189, 240, doi: 10.1088/0067-0049/189/1/240

Davis, A., Jones, S., & Herwig, F. 2019, MNRAS, 484, 3921, doi: 10.1093/mnras/sty3415

de Mink, S. E., Cantiello, M., Langer, N., et al. 2009, A&A, 497, 243, doi: 10.1051/0004-6361/200811439

de Mink, S. E., & Mandel, I. 2016, MNRAS, 460, 3545, doi: 10.1093/mnras/stw1219

deBoer, R. J., Görres, J., Wiescher, M., et al. 2017, Reviews of Modern Physics, 89, 035007, doi: 10.1103/RevModPhys.89.035007

den Hartogh, J. W., Eggenberger, P., & Deheuvels, S. 2020, A&A, 634, L16, doi: 10.1051/0004-6361/202037568

Dufton, P. L., Evans, C. J., Hunter, I., Lennon, D. J., & Schneider, F. R. N. 2019, A&A, 626, A50, doi: 10.1051/0004-6361/201935415

Ertl, T., Janka, H.-T., Woosley, S. E., Sukhbold, T., & Ugliano, M. 2016, ApJ, 818, 124, doi: 10.3847/0004-637X/818/2/124

Ertl, T., Woosley, S., Sukhbold, T., & Janka, H. T. 2020, ApJ, 890, 51, doi: 10.3847/1538-4357/ab6458

Evans, C. J., Taylor, W. D., Hénault-Brunet, V., et al. 2011, A&A, 530, A108, doi: 10.1051/0004-6361/201116782

Farag, E., Timmes, F. X., Taylor, M., Patton, K. M., & Farmer, R. 2020, ApJ, 893, 133, doi: 10.3847/1538-4357/ab7f2c

Farmer, R., Fields, C. E., Petermann, I., et al. 2016, ApJS, 227, 22, doi: 10.3847/1538-4365/227/2/22

Farmer, R., Laplace, E., de Mink, S. E., & Justham, S. 2021, ApJ, 923, 214, doi: 10.3847/1538-4357/ac2f44

Farmer, R., Laplace, E., Ma, J.-z., de Mink, S. E., & Justham, S. 2023, ApJ, 948, 111, doi: 10.3847/1538-4357/acc315

Farmer, R., Renzo, M., de Mink, S. E., Marchant, P., & Justham, S. 2019, ApJ, 887, 53, doi: 10.3847/1538-4357/ab518b

Fields, C. E. 2022, ApJL, 924, L15, doi: 10.3847/2041-8213/ac460c

Fields, C. E., Timmes, F. X., Farmer, R., et al. 2018, ApJS, 234, 19, doi: 10.3847/1538-4365/aaa29b

Friend, D. B., & Abbott, D. C. 1986, ApJ, 311, 701, doi: 10.1086/164809

Fuller, G. M., Fowler, W. A., & Newman, M. J. 1985, ApJ, 293, 1, doi: 10.1086/163208

Fuller, J. 2017, MNRAS, 470, 1642, doi: 10.1093/mnras/stx1314

Fuller, J., & Lu, W. 2022, MNRAS, 511, 3951, doi: 10.1093/mnras/stac317

Fuller, J., Piro, A. L., & Jermyn, A. S. 2019, MNRAS, 485, 3661, doi: 10.1093/mnras/stz514

Fuller, J., & Ro, S. 2018, MNRAS, 476, 1853, doi: 10.1093/mnras/sty369

Galán-Diéguez, D., Berlanas, S. R., Herrero, A., et al. 2026, A&A, 706, A59, doi: 10.1051/0004-6361/202557446

Gilkis, A., Laplace, E., Arcavi, I., Shenar, T., & Schneider, F. R. N. 2025, MNRAS, 540, 3094, doi: 10.1093/mnras/staf884

Gilkis, A., Laplace, E., Drout, M., et al. 2026, arXiv e-prints, arXiv:2604.12868, doi: 10.48550/arXiv.2604.12868

Goldberg, J. A., Bildsten, L., & Paxton, B. 2019, ApJ, 879, 3, doi: 10.3847/1538-4357/ab22b6

Goldberg, J. A., Jiang, Y.-F., & Bildsten, L. 2022, ApJ, 929, 156, doi: 10.3847/1538-4357/ac5ab3

Gommers, R., Virtanen, P., Burovski, E., et al. 2023, scipy/scipy: SciPy 1.11.0, v1.11.0 Zenodo, doi: 10.5281/zenodo.8079889

Gottlieb, O. 2025, ApJL, 992, L3, doi: 10.3847/2041-8213/ae09af

Gottlieb, O., Metzger, B. D., Issa, D., et al. 2025, ApJL, 993, L54, doi: 10.3847/2041-8213/ae0d81

Gottlieb, O., Renzo, M., Metzger, B. D., Goldberg, J. A., & Cantiello, M. 2024, ApJL, 976, L13, doi: 10.3847/2041-8213/ad8563

Grichener, A., Renzo, M., Kerzendorf, W. E., et al. 2025, ApJS, 279, 49, doi: 10.3847/1538-4365/ade717

Griffiths, A., Aloy, M.-Á., & Obergaulinger, M. 2026, arXiv e-prints, arXiv:2605.22927, doi: 10.48550/arXiv.2605.22927

Griffiths, A., Aloy, M.-Á., Hirschi, R., et al. 2025, A&A, 693, A93, doi: 10.1051/0004-6361/202451816

Hamann, W.-R., Koesterke, L., & Wessolowski, U. 1995, A&A, 299, 151

Harris, C. R., Millman, K. J., van der Walt, S. J., et al. 2020, Nature, 585, 357, doi: 10.1038/s41586-020-2649-2

Heger, A., Jeannin, L., Langer, N., & Baraffe, I. 1997,

Heger, A., Langer, N., & Woosley, S. 2000, \apj, 528, 368

Heger, A., Woosley, S. E., & Spruit, H. C. 2005, ApJ, 626, 350, doi: 10.1086/429868

Henneco, J., & Bowman, D. M. 2026, arXiv e-prints, arXiv:2606.13567, doi: 10.48550/arXiv.2606.13567

Henyey, L., Vardya, M. S., & Bodenheimer, P. 1965,

Hirschi, R., Meynet, G., & Maeder, A. 2004, A&A, 425, 649, doi: 10.1051/0004-6361:20041095

Hirschi, R., Meynet, G., & Maeder, A. 2005a, A&A, 433, 1013, doi: 10.1051/0004-6361:20041554

Hirschi, R., Meynet, G., & Maeder, A. 2005b, A&A, 443, 581, doi: 10.1051/0004-6361:20053329

Hirschi, R., Meynet, G., & Maeder, A. 2005c, A&A, 433, 1013, doi: 10.1051/0004-6361:20041554

Hix, W. R., & Thielemann, F.-K. 1993, in American Astronomical Society Meeting Abstracts, Vol. 183, American Astronomical Society Meeting Abstracts, 111.05

Hix, W. R., & Thielemann, F.-K. 1996, ApJ, 460, 869, doi: 10.1086/177016

Hix, W. R., & Thielemann, F.-K. 1999, ApJ, 511, 862, doi: 10.1086/306692

Hunter, I., Lennon, D. J., Dufton, P. L., et al. 2008, A&A, 479, 541, doi: 10.1051/0004-6361:20078511

Hunter, J. D. 2007, Computing in Science & Engineering, 9, 90, doi: 10.1109/MCSE.2007.55

Iglesias, C. A., & Rogers, F. J. 1993, ApJ, 412, 752, doi: 10.1086/172958

Iglesias, C. A., & Rogers, F. J. 1996, ApJ, 464, 943, doi: 10.1086/177381

Issa, J., & Herwig, F. 2025, arXiv e-prints, arXiv:2512.17705, doi: 10.48550/arXiv.2512.17705

Issa, J., Herwig, F., Denissenkov, P. A., & Pignatari, M. 2026, ApJ, 997, 41, doi: 10.3847/1538-4357/ae1efb

Itoh, N., Hayashi, H., Nishikawa, A., & Kohyama, Y. 1996, ApJS, 102, 411, doi: 10.1086/192264

Ivezić, Ž., Kahn, S. M., Tyson, J. A., et al. 2019, ApJ, 873, 111, doi: 10.3847/1538-4357/ab042c

Janka, H.-T. 2012, Annual Review of Nuclear and Particle Science, 62, 407, doi: 10.1146/annurev-nucl-102711-094901

Jermyn, A. S., Anders, E. H., Lecoanet, D., & Cantiello, M. 2022, ApJS, 262, 19, doi: 10.3847/1538-4365/ac7cee

Jermyn, A. S., Schwab, J., Bauer, E., Timmes, F. X., & Potekhin, A. Y. 2021, ApJ, 913, 72, doi: 10.3847/1538-4357/abf48e

Jermyn, A. S., Bauer, E. B., Schwab, J., et al. 2023, ApJS, 265, 15, doi: 10.3847/1538-4365/acae8d

Johnston, C., Michielsen, M., Anders, E. H., et al. 2024, ApJ, 964, 170, doi: 10.3847/1538-4357/ad2343

Jones, S., Hirschi, R., & Nomoto, K. 2014, ApJ, 797, 83, doi: 10.1088/0004-637X/797/2/83

Josiek, J., Ekström, S., & Sander, A. A. C. 2024, A&A, 688, A71, doi: 10.1051/0004-6361/202449281

Joss, P. C., Salpeter, E. E., & Ostriker, J. P. 1973, ApJ, 181, 429, doi: 10.1086/152060

Joyce, M., & Chaboyer, B. 2018, ApJ, 856, 10, doi: 10.3847/1538-4357/aab200

Joyce, M., & Tayar, J. 2023, Galaxies, 11, 75, doi: 10.3390/galaxies11030075

Kasen, D., & Bildsten, L. 2010, ApJ, 717, 245, doi: 10.1088/0004-637X/717/1/245

Kippenhahn, R., Weigert, A., & Weiss, A. 2013, Stellar Structure and Evolution (Springer-Verlag), doi: 10.1007/978-3-642-30304-3

Kunz, R., Fey, M., Jaeger, M., et al. 2002, ApJ, 567, 643, doi: 10.1086/338384

Kuroda, T., Kotake, K., Takiwaki, T., & Thielemann, F.-K. 2018, https://arxiv.org/abs/1801.01293

Landri, C., Ricker, P. M., Renzo, M., Rau, S., & Vigna-Gómez, A. 2025, ApJ, 979, 57, doi: 10.3847/1538-4357/ad9d3c

Langanke, K., & Martínez-Pinedo, G. 2000, Nuclear Physics A, 673, 481, doi: 10.1016/S0375-9474(00)00131-7

Langer, N. 1998, A&A, 329, 551

Laplace, E., Justham, S., Renzo, M., et al. 2021, A&A, 656, A58, doi: 10.1051/0004-6361/202140506

Laplace, E., Schneider, F. R. N., & Podsiadlowski, P. 2025, A&A, 695, A71, doi: 10.1051/0004-6361/202451077

Ledoux, P. 1947, ApJ, 105, 305, doi: 10.1086/144905

Lennon, D. J., Berlanas, S. R., Herrero, A., et al. 2026, A&A, 707, A204, doi: 10.1051/0004-6361/202558539

Limongi, M., & Chieffi, A. 2003, ApJ, 592, 404, doi: 10.1086/375703

Limongi, M., & Chieffi, A. 2018, ApJS, 237, 13, doi: 10.3847/1538-4365/aacb24

Limongi, M., & Chieffi, A. 2020, ApJ, 902, 95, doi: 10.3847/1538-4357/abb4e8

Limongi, M., Roberti, L., Chieffi, A., & Nomoto, K. 2024, ApJS, 270, 29, doi: 10.3847/1538-4365/ad12c1

Limongi, M., Roberti, L., Falla, A., Chieffi, A., & Nomoto, K. 2025, ApJS, 279, 41, doi: 10.3847/1538-4365/addf34

Ma, J.-Z., Farmer, R., de Mink, S. E., & Laplace, E. 2025a, A&A, 704, A179, doi: 10.1051/0004-6361/202555350

Ma, J.-Z., Justham, S., Pakmor, R., et al. 2025b, arXiv e-prints, arXiv:2510.14875, doi: 10.48550/arXiv.2510.14875

MacFadyen, A. I., & Woosley, S. E. 1999, ApJ, 524, 262, doi: 10.1086/307790

Maeder, A., & Meynet, G. 2000, ARA&A, 38, 143, doi: 10.1146/annurev.astro.38.1.143

Mahy, L., Almeida, L. A., Sana, H., et al. 2020, A&A, 634, A119, doi: 10.1051/0004-6361/201936152

Maltsev, K., Schneider, F. R. N., Mandel, I., et al. 2025, A&A, 700, A20, doi: 10.1051/0004-6361/202554931

Mandel, I., & Broekgaarden, F. S. 2022, Living Reviews in Relativity, 25, 1, doi: 10.1007/s41114-021-00034-3

Mandel, I., & de Mink, S. E. 2016, MNRAS, 458, 2634, doi: 10.1093/mnras/stw379

Marchant, P., Langer, N., Podsiadlowski, P., Tauris, T. M., & Moriya, T. J. 2016, A&A, 588, A50, doi: 10.1051/0004-6361/201628133

Marchant, P., Renzo, M., Farmer, R., et al. 2019, ApJ, 882, 36, doi: 10.3847/1538-4357/ab3426

Martins, F., Depagne, E., Russeil, D., & Mahy, L. 2013, A&A, 554, A23, doi: 10.1051/0004-6361/201321282

Martins, F., Hillier, D. J., Bouret, J. C., et al. 2009, A&A, 495, 257, doi: 10.1051/0004-6361:200811014

Melson, T., Janka, H.-T., & Marek, A. 2015, ApJL, 801, L24, doi: 10.1088/2041-8205/801/2/L24

Mezzacappa, A. 2026, arXiv e-prints, arXiv:2604.24970, doi: 10.48550/arXiv.2604.24970

Mombarg, J. S. G., Rieutord, M., & Espinosa Lara, F. 2024, A&A, 683, A94, doi: 10.1051/0004-6361/202348466

Morozova, V., Piro, A. L., Renzo, M., et al. 2015, ApJ, 814, 63, doi: 10.1088/0004-637X/814/1/63

Müller, B. 2016, PASA, 33, e048, doi: 10.1017/pasa.2016.40

Müller, B., Heger, A., Liptai, D., & Cameron, J. B. 2016, MNRAS, 460, 742, doi: 10.1093/mnras/stw1083

Müller, B., Melson, T., Heger, A., & Janka, H.-T. 2017, MNRAS, 472, 491, doi: 10.1093/mnras/stx1962

Myers, M. A., Campbell, C. B., Patton, K. M., et al. 2026, arXiv e-prints, arXiv:2604.22605, doi: 10.48550/arXiv.2604.22605

Nakamura, K., Takiwaki, T., Matsumoto, J., & Kotake, K. 2025, MNRAS, 536, 280, doi: 10.1093/mnras/stae2611

Nathaniel, K., Vigna-Gómez, A., Grichener, A., et al. 2025, A&A, 694, A83, doi: 10.1051/0004-6361/202451531

Nazé, Y., Britavskiy, N., Rauw, G., Labadie-Bartz, J., & Simón-Díaz, S. 2023, MNRAS, 525, 1641, doi: 10.1093/mnras/stad2280

Nazé, Y., Rauw, G., Kołaczek-Szymański, P. A., Britavskiy, N., & Labadie-Bartz, J. 2025, A&A, 703, A239, doi: 10.1051/0004-6361/202556441

Nieuwenhuijzen, H., & de Jager, C. 1990, \aap, 231, 134

Noll, A., Basu, S., & Hekker, S. 2025, A&A, 704, A188, doi: 10.1051/0004-6361/202554393

Nomoto, K. 1984, in Astrophysics and Space Science Library, Vol. 109, Stellar Nucleosynthesis, ed. C. Chiosi & A. Renzini, 239, doi: 10.1007/978-94-009-6348-1_20

Nomoto, K., & Hashimoto, M. 1986, in Bulletin of the American Astronomical Society, Vol. 18, 635

Nugis, T., & Lamers, H. J. G. L. M. 2000, A&A, 360, 227

O'Connor, E., & Ott, C. D. 2010, Classical and Quantum Gravity, 27, 114103, doi: 10.1088/0264-9381/27/11/114103

O'Connor, E., & Ott, C. D. 2011, ApJ, 730, 70

Oda, T., Hino, M., Muto, K., Takahara, M., & Sato, K. 1994, Atomic Data and Nuclear Data Tables, 56, 231, doi: 10.1006/adnd.1994.1007

Ott, C. D., Roberts, L. F., da Silva Schneider, A., et al. 2018, ApJL, 855, L3, doi: 10.3847/2041-8213/aaa967

Patton, K. M., Lunardini, C., & Farmer, R. J. 2017, ApJ, 840, 2, doi: 10.3847/1538-4357/aa6ba8

Paxton, B., Bildsten, L., Dotter, A., et al. 2011, ApJS, 192, 3, doi: 10.1088/0067-0049/192/1/3

Paxton, B., Cantiello, M., Arras, P., et al. 2013, ApJS, 208, 4, doi: 10.1088/0067-0049/208/1/4

Paxton, B., Marchant, P., Schwab, J., et al. 2015, ApJS, 220, 15, doi: 10.1088/0067-0049/220/1/15

Paxton, B., Marchant, P., Schwab, J., et al. 2016, ApJS, 223, 18, doi: 10.3847/0067-0049/223/1/18

Paxton, B., Schwab, J., Bauer, E. B., et al. 2018, ApJS, 234, 34, doi: 10.3847/1538-4365/aaa5a8

Paxton, B., Smolec, R., Schwab, J., et al. 2019, ApJS, 243, 10, doi: 10.3847/1538-4365/ab2241

Perna, R., Lazzati, D., & Cantiello, M. 2018, ApJ, 859, 48, doi: 10.3847/1538-4357/aabcc1

Pols, O. R., Tout, C. A., Eggleton, P. P., & Han, Z. 1995,

Puls, J., Vink, J. S., & Najarro, F. 2008, A&A Rv, 16, 209, doi: 10.1007/s00159-008-0015-8

Quataert, E., Lecoanet, D., & Coughlin, E. R. 2019, MNRAS, 485, L83, doi: 10.1093/mnrasl/slz031

Rakavy, G., & Shaviv, G. 1967, ApJ, 148, 803, doi: 10.1086/149204

Ramírez-Agudelo, O. H., Simón-Díaz, S., Sana, H., et al. 2013, A&A, 560, A29, doi: 10.1051/0004-6361/201321986

Ramírez-Agudelo, O. H., Sana, H., de Mink, S. E., et al. 2015, A&A, 580, A92, doi: 10.1051/0004-6361/201425424

Renzo, M. 2024, MESA files for: "Progenitor stars calculated with small reaction networks should not be used as initial conditions for core collapse", Zenodo, doi: 10.5281/ZENODO.11375522

Renzo, M., Cantiello, M., Metzger, B. D., & Jiang, Y.-F. 2020, ApJL, 904, L13, doi: 10.3847/2041-8213/abc6a6

Renzo, M., Goldberg, J. A., Grichener, A., Gottlieb, O., & Cantiello, M. 2024, Research Notes of the American Astronomical Society, 8, 152, doi: 10.3847/2515-5172/ad530e

Renzo, M., & Götberg, Y. 2021, ApJ, 923, 277, doi: 10.3847/1538-4357/ac29c5

Renzo, M., Ott, C. D., Shore, S. N., & de Mink, S. E. 2017, A&A, 603, A118, doi: 10.1051/0004-6361/201730698

Renzo, M., Zapartas, E., Justham, S., et al. 2023, ApJL, 942, L32, doi: 10.3847/2041-8213/aca4d3

Rieutord, M., Brossier-Sécher, E., & Mombarg, J. S. G. 2025, A&A, 699, A146, doi: 10.1051/0004-6361/202555003

Rizzuti, F., Hirschi, R., Arnett, W. D., et al. 2023, Monthly Notices of the Royal Astronomical Society, stad1572, doi: 10.1093/mnras/stad1572

Romagnolo, A., Broekgaarden, F. S., Antoniadis, K., & Gormaz-Matamala, A. C. 2026, arXiv e-prints, arXiv:2601.02263, doi: 10.48550/arXiv.2601.02263

Schneider, F. R. N., Podsiadlowski, P., & Laplace, E. 2024, A&A, 686, A45, doi: 10.1051/0004-6361/202347854

Schneider, F. R. N., Podsiadlowski, P., & Müller, B. 2021, A&A, 645, A5, doi: 10.1051/0004-6361/202039219

Schultz, W. C., Bildsten, L., & Jiang, Y.-F. 2020, ApJ, 902, 67, doi: 10.3847/1538-4357/abb405

Sen, K., Olejak, A., & Banerjee, S. 2025, A&A, 696, A54, doi: 10.1051/0004-6361/202553829

Shah, N., Renzo, M., Sen, K., Grichener, A., & Breivik, K. 2026, arXiv e-prints, arXiv:2602.12327, doi: 10.48550/arXiv.2602.12327

Sharpe, K., van Son, L. A. C., de Mink, S. E., et al. 2024, ApJ, 966, 9, doi: 10.3847/1538-4357/ad2f3e

Shen, Y., Guo, B., deBoer, R. J., et al. 2023, ApJ, 945, 41, doi: 10.3847/1538-4357/acb7de

Shenar, T., Bodensteiner, J., Sana, H., et al. 2024, A&A, 690, A289, doi: 10.1051/0004-6361/202451586

Shishkin, D., & Soker, N. 2021, MNRAS, 508, L43, doi: 10.1093/mnrasl/slab105

Siegel, D. M., Agarwal, A., Barnes, J., et al. 2022, ApJ, 941, 100, doi: 10.3847/1538-4357/ac8d04

Skoutnev, V. A., & Beloborodov, A. M. 2024, ApJ, 974, 290, doi: 10.3847/1538-4357/ad71c8

Skoutnev, V. A., & Beloborodov, A. M. 2025, ApJ, 988, 195, doi: 10.3847/1538-4357/ade547

Smith, N. 2014, \araa

Soker, N. 2024, The Open Journal of Astrophysics, 7, 31, doi: 10.33232/001c.117147

Spruit, H. C. 1999, A&A, 349, 189, doi: 10.48550/arXiv.astro-ph/9907138

Spruit, H. C. 2002, A&A, 381, 923, doi: 10.1051/0004-6361:20011465

Stuck, M., Pratt, J., Baraffe, I., et al. 2025, A&A, 698, A304, doi: 10.1051/0004-6361/202555172

Sukhbold, T., & Woosley, S. E. 2016, ApJL, 820, L38, doi: 10.3847/2041-8205/820/2/L38

Sukhbold, T., Woosley, S. E., & Heger, A. 2018, ApJ, 860, 93, doi: 10.3847/1538-4357/aac2da

Takahashi, K. 2018, ApJ, 863, 153, doi: 10.3847/1538-4357/aad2d2

Takahashi, K., Yoshida, T., & Umeda, H. 2013, ApJ, 771, 28, doi: 10.1088/0004-637X/771/1/28

Takiwaki, T., Kotake, K., & Suwa, Y. 2016, MNRAS, 461, L112, doi: 10.1093/mnrasl/slw105

The LIGO Scientific Collaboration, the Virgo Collaboration, & the KAGRA Collaboration. 2026, arXiv e-prints, arXiv:2605.27226, doi: 10.48550/arXiv.2605.27226

Thielemann, F.-K., Nomoto, K., & Hashimoto, M.-A. 1996, ApJ, 460, 408, doi: 10.1086/176980

Timmes, F. X., Hoffman, R. D., & Woosley, S. E. 2000, ApJS, 129, 377, doi: 10.1086/313407

Tong, H., Fishbach, M., Thrane, E., et al. 2026, Nature, 652, 874, doi: 10.1038/s41586-026-10359-0

Ugliano, M., Janka, H.-T., Marek, A., & Arcones, A. 2012, ApJ, 757, 69, doi: 10.1088/0004-637X/757/1/69

Van Rossum, G., & Drake, F. L. 2009, Python 3 Reference Manual (Scotts Valley, CA: CreateSpace)

Vartanyan, D., Burrows, A., Radice, D., Skinner, M. A., & Dolence, J. 2019, MNRAS, 482, 351, doi: 10.1093/mnras/sty2585

Vartanyan, D., Laplace, E., Renzo, M., et al. 2021, ApJL, 916, L5, doi: 10.3847/2041-8213/ac0b42

Vink, J., de Koter, A., & Lamers, H. 2000, \aap, 362, 295

Vink, J. S. 2015, in Astrophysics and Space Science Library, Vol. 412, Very Massive Stars in the Local Universe, ed. J. S. Vink, 77, doi: 10.1007/978-3-319-09596-7_4

Vink, J. S., de Koter, A., & Lamers, H. J. G. L. M. 2001, A&A, 369, 574, doi: 10.1051/0004-6361:20010127

Virtanen, P., Gommers, R., Oliphant, T. E., et al. 2020, Nature Methods, 17, 261, doi: 10.1038/s41592-019-0686-2

Wagg, T., Broekgaarden, F., Van-Lane, P., Wu, K., & Gültekin, K. 2025, TomWagg/software-citation-station: v1.4, v1.4 Zenodo, doi: 10.5281/zenodo.17654855

Wagg, T., & Broekgaarden, F. S. 2024, arXiv e-prints, arXiv:2406.04405. https://arxiv.org/abs/2406.04405

Wagg, T., Johnston, C., Bellinger, E. P., et al. 2024, arXiv e-prints, arXiv:2403.05627, doi: 10.48550/arXiv.2403.05627

Weaver, T. A., Zimmerman, G. B., & Woosley, S. E. 1978, ApJ, 225, 1021, doi: 10.1086/156569

Woosley, S. E. 1993, ApJ, 405, 273, doi: 10.1086/172359

Woosley, S. E. 2007, Nature Physics, 3, 832, doi: 10.1038/nphys804

Woosley, S. E., & Heger, A. 2006, ApJ, 637, 914, doi: 10.1086/498500

Woosley, S. E., & Heger, A. 2021, ApJL, 912, L31, doi: 10.3847/2041-8213/abf2c4

Woosley, S. E., Heger, A., & Weaver, T. A. 2002, Reviews of Modern Physics, 74, 1015, doi: 10.1103/RevModPhys.74.1015

Xin, C., Renzo, M., & Metzger, B. D. 2022, Monthly Notices of the Royal Astronomical Society, 516, 5816, doi: 10.1093/mnras/stac2551

Yadav, N., Müller, B., Janka, H. T., Melson, T., & Heger, A. 2020, ApJ, 890, 94, doi: 10.3847/1538-4357/ab66bb

Yoon, S.-C., Langer, N., & Norman, C. 2006, A&A, 460, 199, doi: 10.1051/0004-6361:20065912

Yoshida, T., Umeda, H., & Nomoto, K. 2005, ApJ, 631, 1039, doi: 10.1086/432654

Zha, S., Leung, S.-C., Suzuki, T., & Nomoto, K. 2019, ApJ, 886, 22, doi: 10.3847/1538-4357/ab4b4b

# APPENDIX

## A. MESA_128.NET

Fig. 6 shows these isotopes (in blue) with the connecting reaction links in `mesa_128.net` and for comparison, the 22-isotope list and reaction links of `approx21_cr60_plus_co56.net` used for our small nuclear network model (in pink), which includes also $^{60}$Cr (not included in `mesa_128.net`, red outline), as an example of isotope that can be included to tweak compound weak reactions.

The list of isotopes included in `mesa_128.net` is: $n$, $^{1-2}$H, $^{2}$H, $^{3}$He, $^{4}$He, $^{7}$Li, $^{7}$Be, $^{9}$Be, $^{10}$Be, $^{8}$B, $^{12}$C, $^{13}$C, $^{13}$N, $^{14}$N, $^{15}$N, $^{14}$O, $^{15}$O, $^{16}$O, $^{17}$O, $^{18}$O, $^{17}$F, $^{18}$F, $^{19}$F, $^{18}$Ne, $^{19}$Ne, $^{20}$Ne, $^{21}$Ne, $^{22}$Ne, $^{21}$Na, $^{22}$Na, $^{23}$Na, $^{24}$Na, $^{23}$Mg, $^{24}$Mg, $^{25}$Mg, $^{26}$Mg, $^{25}$Al, $^{26}$Al, $^{27}$Al, $^{28}$Al, $^{27}$Si, $^{28}$Si, $^{29}$Si, $^{30}$Si, $^{30}$P, $^{31}$P, $^{32}$P, $^{31}$S, $^{32}$S, $^{33}$S, $^{34}$S, $^{35}$Cl, $^{36}$Cl, $^{37}$Cl, $^{35}$Ar, $^{36}$Ar, $^{37}$Ar, $^{38}$Ar, $^{39}$K, $^{40}$K, $^{41}$K, $^{42}$K, $^{43}$K, $^{39}$Ca, $^{40}$Ca, $^{41}$Ca, $^{42}$Ca, $^{43}$Ca, $^{44}$Ca, $^{43}$Sc, $^{44}$Sc, $^{45}$Sc, $^{46}$Sc, $^{44}$Ti, $^{45}$Ti, $^{46}$Ti, $^{47}$Ti, $^{48}$Ti, $^{47}$V, $^{48}$V, $^{49}$V, $^{50}$V, $^{51}$V, $^{48}$Cr, $^{49}$Cr, $^{50}$Cr, $^{51}$Cr, $^{52}$Cr, $^{53}$Cr, $^{54}$Cr, $^{55}$Cr, $^{56}$Cr, $^{57}$Cr, $^{51}$Mn, $^{52}$Mn, $^{53}$Mn, $^{54}$Mn, $^{55}$Mn, $^{56}$Mn, $^{52}$Fe, $^{53}$Fe, $^{54}$Fe, $^{55}$Fe, $^{56}$Fe, $^{57}$Fe, $^{58}$Fe, $^{55}$Co, $^{56}$Co, $^{57}$Co, $^{58}$Co, $^{59}$Co, $^{60}$Co, $^{55}$Ni, $^{56}$Ni, $^{57}$Ni, $^{58}$Ni, $^{59}$Ni, $^{60}$Ni, $^{61}$Ni, $^{59}$Cu, $^{60}$Cu, $^{61}$Cu, $^{62}$Cu, $^{60}$Zn, $^{61}$Zn, $^{62}$Zn, $^{63}$Zn, $^{64}$Zn.

## B. EXPLODABILITY METRICS AND SURFACE COMPOSITION

Tab. 1 lists for each successful model in our grid a few key quantities related to their "explodability" and their surface composition. Specifically, we list $\xi_{1.75}$ and $M_{\rm Fe}$ shown also in Fig. 5, $M_4$ is the outermost mass coordinate where the specific entropy is $s = 4\,k_B N_A$ and the mass-gradient at that location $\mu_4$, which together can be combined in the two-parameter explodability criterion by T. Ertl et al. (2016); T. Ertl et al. (2020).

The last three columns provide the surface mass fractions of the most abundant elements, which are Helium $Y_{\rm surf} \equiv X_{\rm surf}(^4{\rm He})$, Carbon, and Oxygen. These do not necessarily sum to 1 because of other isotopes present in our calculations, and are presumably sensitive to the wind mass-loss rates and mixing processes. Interestingly, most of our models still retain a significant amount of helium at the surface (plus, some helium-rich material can appear in the core very late due to photodisintegration of iron-group nuclei).

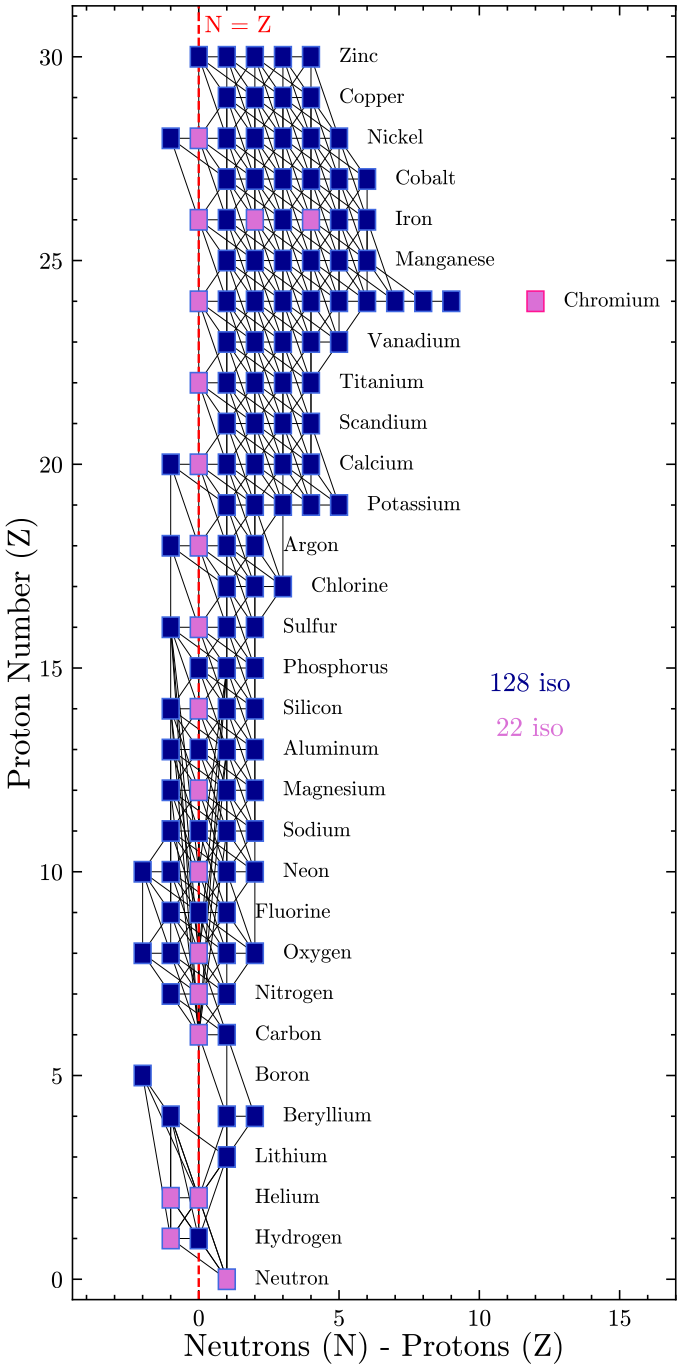



**Figure 6.** Isotope list and reaction links for the 128-isotope network used in this study and an example of 22-isotope small network.

## C. DESCRIPTION OF THE DATA PRODUCTS

We share the stellar models described in this paper at doi.org/10.5281/zenodo.14286306, to provide ini-

tial conditions for stellar explosion studies. In the repository, `template.tar.xz` contains the input files for our MESA simulations, designed for version `r24.03.1`. This includes a customized `rn` script that tries to use the `mail` command-line utility to send the last MESA `pgstar` output and the last lines of the terminal output to the user. This requires modifying the email address, and may not work depending on the setup of `mail` on the machine running the models. Failure to send the email notification will not affect the MESA calculations in any way.

The numerical results are in `grid.tar`, which contains a separate tarball for each star, labeled as `<M>_rot<O>.tar.xz` where `<M>` is the ZAMS mass in $M_\odot$ with one decimal digit, and `<O>` is the initial value of $\omega_{\rm ZAMS}/\omega_{\rm crit}$ with two decimal digits. We only include the models reaching the onset of core collapse successfully ($v_{\rm infall} \lesssim 300\,{\rm km\ s^{-1}}$) and pass our visual inspection (cf. Fig. 1).

Each `<M>_rot<O>.tar.xz` provides a MESA work directory containing the input files for that specific model, the full terminal output of the run (`output.txt`), a few plots generated on-the-fly by MESA `pgstar` inside the subfolder `png`, the `history.data` file and selected `profile*.data` in the subfolder `LOGS`. All profiles at the onset of core collapse are named `CHE_single_core_collapse.data`, we also provide profiles and MESA `*.mod` files for the first time the temperature in the star exceeds $\log_{10}(T/[{\rm K}]) = 8.95$ (`CHE_logT895.mod`), core Oxygen depletion (`O_depl.mod`, defined as the first timestep when the central mass fractions $X_c(^{16}{\rm O}) < 0.1$ and $X_c(^{12}{\rm C})$ and $X_c(^4{\rm He})$ are both $< 0.001$ and $X_c(^1{\rm H}) < 0.5$) and core Silicon depletion (`Si_depl.mod`, defined as the first timestep when the central mass fractions $X_c(^{28}{\rm Si}), X_c(^{16}{\rm O}), X_c(^{12}{\rm C}) < 5\times10^{-3}$ and $X_c(^4{\rm He}) < 0.2$ and $X_c(^1{\rm H}) < 0.5$). These conditions conservatively and consistently define the moment of core Oxygen and Silicon core depletion across the grid with 128 isotopes.

Profiles and/or `*.mod` files can be used to initialize multi-dimensional simulations at times earlier than the onset of core collapse, for example, to study pre-collapse shell mergers and/or initializing self-consistent (non-MLT) convection velocity profiles (e.g., B. Côté et al. 2020; N. Yadav et al. 2020; C. E. Fields 2022; J. Issa & F. Herwig 2025; A. Griffiths et al. 2026; J. Issa et al. 2026)

The scripts used for the setup and analysis of our MESA models are in `scripts.tar.xz`, which also contains specifications for the dependencies. All figures can be reproduced starting with the computed MESA results in `grid.tar` (and the zenodo **dataset (M. Renzo 2024) from M. Renzo et al. 2024 for the 22-isotope model available at doi.org/10.5281/zenodo.11375522**).

Finally, the (interactive) `build.sh` script allows for full reproduction of this manuscript, including downloading of the data from their `doi`. An evolving version of the code is available also at github.com/mathren/CHE_lGRB_progenitors.

**Table 1**. **Core masses**, explodability metrics, and final surface composition for our CHE models. The first two columns give the initial ZAMS values of mass and initial rotational frequency in units of $\omega_{\rm crit}$. The following three columns give the final iron ($M_{\rm Fe}$), silicon ($M_{\rm Si}$), and carbon-oxygen core masses ($M_{\rm CO}$), defined as the outermost location where the mass fraction of an element (or their sum in the case of carbon and oxygen) rises above 0.1 and the mass fraction of the lighter element drops below 0.1. For the iron core, we consider $\sum_j X_j > 0.1$ where the sum runs over all species with $A_j > 46$. The next two columns gives the Helium core mass $M_{\rm He}$ and total mass $M$ at terminal age main sequence, defined as $X_c(^1{\rm H}) \leq 10^{-2}$. Then we list the compactness parameter $\xi_{1.75}$ (E. O'Connor & C. D. Ott 2011) calculated at mass coordinate $\mathcal{M} = 1.75\,M_\odot$ (A. Burrows et al. 2024), and the two parameters $M_4 \equiv \max(m(s < 4\,k_B N_A))$ and $\mu_4 = dm/dr(m = M_4)$ for the T. Ertl et al. (2016); T. Ertl et al. (2020) explodability criterion. The last three columns are the final surface mass fractions of helium $Y_{\rm surf} \equiv X_{\rm surf}(^4{\rm He})$, carbon, and oxygen. These do not necessarily sum to 1 because of other isotopes present in our calculations.

| $M_{\rm ZAMS}$ $[M_\odot]$ | $\omega_{\rm ZAMS}$ $[\omega_{\rm crit}]$ | $M_{\rm Fe}$ $[M_\odot]$ | $M_{\rm Si}$ $[M_\odot]$ | $M_{\rm CO}$ $[M_\odot]$ | $M_{\rm He,TAMS}$ $[M_\odot]$ | $M_{\rm TAMS}$ $[M_\odot]$ | $\xi_{1.75}$ | $M_4$ $[M_\odot]$ | $\mu_4$ $[M_\odot/10^3\,{\rm km}]$ | $Y_{\rm surf}$ | $X_{\rm surf}(^{12}{\rm C})$ | $X_{\rm surf}(^{16}{\rm O})$ |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 30.00 | 0.55 | 1.41 | 1.41 | 19.15 | 19.20 | 24.07 | 0.592 | 1.70 | 0.173 | 0.32 | 0.16 | 0.48 |
| 30.00 | 0.94 | 3.14 | 1.41 | 18.37 | 17.87 | 23.01 | 0.395 | 1.56 | 0.114 | 0.26 | 0.16 | 0.51 |

(continued from previous page)

| $M_{\rm ZAMS}$ $[M_\odot]$ | $\omega_{\rm ZAMS}$ $[\omega_{\rm crit}]$ | $M_{\rm Fe}$ $[M_\odot]$ | $M_{\rm Si}$ $[M_\odot]$ | $M_{\rm CO}$ $[M_\odot]$ | $M_{\rm He,TAMS}$ $[M_\odot]$ | $M_{\rm TAMS}$ $[M_\odot]$ | $\xi_{1.75}$ | $M_4$ $[M_\odot]$ | $\mu_4$ $[M_\odot/10^3\,{\rm km}]$ | $Y_{\rm surf}$ | $X_{\rm surf}(^{12}{\rm C})$ | $X_{\rm surf}(^{16}{\rm O})$ |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 32.00 | 0.55 | 2.23 | 1.48 | 20.85 | 21.93 | 26.00 | 0.544 | 1.96 | 0.134 | 0.31 | 0.21 | 0.42 |
| 32.00 | 0.70 | 1.76 | 1.71 | 19.33 | 19.27 | 24.61 | 0.616 | 1.81 | 0.133 | 0.31 | 0.15 | 0.46 |
| 32.00 | 0.84 | 2.21 | 1.34 | 19.66 | 19.14 | 24.30 | 0.468 | 1.58 | 0.143 | 0.46 | 0.09 | 0.34 |
| 34.00 | 0.60 | 2.10 | 1.54 | 21.59 | 20.81 | 26.29 | 0.597 | 1.88 | 0.133 | 0.66 | 0.04 | 0.22 |
| 34.00 | 0.65 | 1.33 | 1.32 | 20.57 | 20.63 | 26.06 | 0.650 | 1.95 | 0.179 | 0.37 | 0.14 | 0.44 |
| 34.00 | 0.74 | 1.45 | 1.45 | 20.53 | 20.50 | 25.67 | 0.648 | 2.05 | 0.166 | 0.24 | 0.17 | 0.54 |
| 34.00 | 0.84 | 1.76 | 1.76 | 20.45 | 20.40 | 25.53 | 0.651 | 1.90 | 0.132 | 0.26 | 0.16 | 0.53 |
| 34.00 | 0.94 | 1.46 | 1.46 | 19.73 | 20.26 | 25.44 | 0.652 | 1.91 | 0.153 | 0.40 | 0.14 | 0.41 |
| 34.00 | 0.99 | 2.11 | 1.49 | 20.62 | 20.30 | 25.41 | 0.545 | 1.85 | 0.127 | 0.73 | 0.04 | 0.17 |
| 36.00 | 0.60 | 2.10 | 1.46 | 22.01 | 22.07 | 27.53 | 0.651 | 1.90 | 0.138 | 0.26 | 0.16 | 0.50 |
| 36.00 | 0.65 | 3.39 | 1.46 | 21.85 | 21.89 | 27.23 | 0.542 | 1.76 | 0.186 | 0.28 | 0.16 | 0.49 |
| 36.00 | 0.70 | 2.11 | 1.52 | 21.69 | 21.83 | 27.05 | 0.683 | 1.92 | 0.144 | 0.25 | 0.17 | 0.50 |
| 36.00 | 0.74 | 1.49 | 1.50 | 21.79 | 21.74 | 26.92 | 0.631 | 2.08 | 0.180 | 0.82 | 0.03 | 0.10 |
| 36.00 | 0.79 | 2.20 | 1.56 | 21.88 | 21.62 | 26.81 | 0.551 | 1.99 | 0.134 | 0.75 | 0.04 | 0.15 |
| 36.00 | 0.84 | 1.31 | 1.31 | 20.93 | 21.56 | 26.67 | 0.640 | 2.14 | 0.193 | 0.29 | 0.17 | 0.51 |
| 36.00 | 0.89 | 1.42 | 1.42 | 21.26 | 21.49 | 26.66 | 0.651 | 2.11 | 0.180 | 0.25 | 0.17 | 0.53 |
| 36.00 | 0.94 | 2.13 | 1.51 | 21.17 | 21.58 | 26.64 | 0.615 | 1.94 | 0.134 | 0.29 | 0.16 | 0.47 |
| 36.00 | 0.99 | 2.12 | 1.45 | 21.32 | 21.56 | 26.57 | 0.530 | 1.91 | 0.131 | 0.52 | 0.07 | 0.30 |
| 38.00 | 0.60 | 2.18 | 1.64 | 23.39 | 23.32 | 28.79 | 0.696 | 2.12 | 0.180 | 0.76 | 0.05 | 0.14 |
| 38.00 | 0.70 | 2.36 | 1.43 | 22.65 | 23.06 | 28.25 | 0.559 | 1.77 | 0.148 | 0.46 | 0.08 | 0.34 |
| 38.00 | 0.74 | 2.45 | 1.45 | 22.81 | 23.04 | 28.09 | 0.536 | 1.69 | 0.150 | 0.61 | 0.07 | 0.23 |
| 38.00 | 0.79 | 1.66 | 1.66 | 22.67 | 22.89 | 28.00 | 0.661 | 2.16 | 0.229 | 0.70 | 0.06 | 0.18 |
| 38.00 | 0.84 | 1.41 | 1.41 | 22.35 | 22.80 | 27.89 | 0.642 | 2.07 | 0.197 | 0.70 | 0.06 | 0.17 |
| 38.00 | 0.89 | 1.54 | 1.54 | 21.96 | 22.80 | 27.84 | 0.604 | 1.74 | 0.150 | 0.53 | 0.10 | 0.30 |
| 38.00 | 0.94 | 1.42 | 1.42 | 22.58 | 22.80 | 27.76 | 0.640 | 1.98 | 0.177 | 0.67 | 0.07 | 0.19 |
| 38.00 | 0.99 | 2.16 | 1.54 | 22.27 | 22.73 | 27.76 | 0.575 | 1.99 | 0.121 | 0.48 | 0.09 | 0.32 |
| 40.00 | 0.55 | 1.48 | 1.48 | 24.38 | 24.74 | 30.46 | 0.588 | 1.81 | 0.204 | 0.28 | 0.17 | 0.50 |
| 40.00 | 0.60 | 1.62 | 1.62 | 24.36 | 24.63 | 29.94 | 0.656 | 2.10 | 0.177 | 0.85 | 0.06 | 0.07 |
| 40.00 | 0.65 | 2.76 | 1.65 | 23.94 | 24.50 | 29.62 | 0.637 | 1.81 | 0.158 | 0.83 | 0.07 | 0.08 |
| 40.00 | 0.70 | 2.23 | 1.51 | 23.58 | 24.38 | 29.39 | 0.637 | 1.94 | 0.135 | 0.45 | 0.10 | 0.33 |
| 40.00 | 0.79 | 2.18 | 1.55 | 23.47 | 24.15 | 29.08 | 0.474 | 1.94 | 0.114 | 0.27 | 0.12 | 0.45 |
| 40.00 | 0.84 | 2.16 | 1.52 | 23.24 | 24.06 | 28.99 | 0.541 | 1.80 | 0.146 | 0.55 | 0.10 | 0.27 |
| 40.00 | 0.89 | 2.68 | 23.82 | 23.34 | 24.08 | 28.96 | 0.609 | 1.61 | 0.218 | 0.90 | 0.07 | 0.03 |
| 40.00 | 0.94 | 1.49 | 1.49 | 23.43 | 23.99 | 28.91 | 0.666 | 2.07 | 0.178 | 0.71 | 0.09 | 0.16 |
| 40.00 | 0.99 | 1.37 | 1.37 | 23.40 | 24.01 | 28.90 | 0.665 | 2.08 | 0.181 | 0.88 | 0.07 | 0.04 |
| 42.00 | 0.60 | 1.72 | 1.72 | 25.12 | 26.00 | 31.06 | 0.698 | 2.00 | 0.186 | 0.73 | 0.11 | 0.14 |
| 42.00 | 0.65 | 1.59 | 1.59 | 24.82 | 25.74 | 30.80 | 0.602 | 2.14 | 0.147 | 0.84 | 0.11 | 0.05 |
| 42.00 | 0.70 | 2.07 | 1.77 | 24.53 | 25.67 | 30.53 | 0.602 | 1.88 | 0.135 | 0.34 | 0.15 | 0.41 |
| 42.00 | 0.74 | 1.81 | 1.81 | 24.57 | 25.48 | 30.44 | 0.684 | 1.88 | 0.178 | 0.74 | 0.11 | 0.12 |
| 42.00 | 0.79 | 1.89 | 1.89 | 24.39 | 25.48 | 30.24 | 0.671 | 2.18 | 0.211 | 0.77 | 0.11 | 0.10 |
| 42.00 | 0.84 | 2.16 | 1.40 | 24.39 | 25.36 | 30.13 | 0.464 | 1.74 | 0.143 | 0.44 | 0.12 | 0.33 |
| 42.00 | 0.89 | 1.58 | 1.58 | 24.16 | 25.29 | 30.03 | 0.642 | 2.04 | 0.197 | 0.78 | 0.11 | 0.09 |
| 42.00 | 0.94 | 2.32 | 1.45 | 24.25 | 25.31 | 30.03 | 0.527 | 1.72 | 0.162 | 0.67 | 0.11 | 0.17 |
| 42.00 | 0.99 | 2.08 | 1.58 | 23.77 | 25.31 | 30.01 | 0.613 | 1.91 | 0.138 | 0.44 | 0.13 | 0.33 |
| 44.00 | 0.60 | 2.18 | 1.50 | 26.07 | 27.27 | 32.31 | 0.549 | 1.77 | 0.150 | 0.37 | 0.15 | 0.38 |
| 44.00 | 0.65 | 2.17 | 1.46 | 25.58 | 27.12 | 31.93 | 0.579 | 1.93 | 0.131 | 0.40 | 0.15 | 0.35 |
| 44.00 | 0.70 | 2.41 | 1.00 | 25.47 | 26.94 | 31.67 | 0.493 | 1.68 | 0.140 | 0.63 | 0.15 | 0.18 |

(continued from previous page)

| $M_{\rm ZAMS}$ $[M_\odot]$ | $\omega_{\rm ZAMS}$ $[\omega_{\rm crit}]$ | $M_{\rm Fe}$ $[M_\odot]$ | $M_{\rm Si}$ $[M_\odot]$ | $M_{\rm CO}$ $[M_\odot]$ | $M_{\rm He,TAMS}$ $[M_\odot]$ | $M_{\rm TAMS}$ $[M_\odot]$ | $\xi_{1.75}$ | $M_4$ $[M_\odot]$ | $\mu_4$ $[M_\odot/10^3\,{\rm km}]$ | $Y_{\rm surf}$ | $X_{\rm surf}(^{12}{\rm C})$ | $X_{\rm surf}(^{16}{\rm O})$ |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 44.00 | 0.74 | 1.37 | 1.37 | 25.31 | 26.79 | 31.51 | 0.645 | 2.07 | 0.190 | 0.79 | 0.15 | 0.06 |
| 44.00 | 0.79 | 2.21 | 1.49 | 25.22 | 26.69 | 31.35 | 0.497 | 1.95 | 0.134 | 0.46 | 0.15 | 0.31 |
| 44.00 | 0.84 | 2.02 | 2.02 | 25.02 | 26.62 | 31.27 | 0.661 | 2.13 | 0.236 | 0.78 | 0.15 | 0.07 |
| 44.00 | 0.89 | 2.77 | 1.33 | 25.17 | 26.54 | 31.16 | 0.674 | 1.74 | 0.205 | 0.69 | 0.15 | 0.14 |
| 44.00 | 0.94 | 1.44 | 1.44 | 25.05 | 26.49 | 31.14 | 0.580 | 1.95 | 0.148 | 0.38 | 0.15 | 0.37 |
| 44.00 | 0.99 | 2.10 | 1.58 | 24.98 | 26.52 | 31.12 | 0.622 | 1.87 | 0.145 | 0.39 | 0.15 | 0.35 |
| 46.00 | 0.55 | 1.75 | 1.75 | 27.21 | 28.91 | 33.86 | 0.531 | 1.94 | 0.141 | 0.53 | 0.17 | 0.24 |
| 46.00 | 0.60 | 1.95 | 1.95 | 26.89 | 28.53 | 33.42 | 0.664 | 2.05 | 0.226 | 0.73 | 0.18 | 0.08 |
| 46.00 | 0.65 | 2.83 | 1.89 | 26.33 | 28.38 | 33.06 | 0.665 | 1.96 | 0.199 | 0.75 | 0.18 | 0.07 |
| 46.00 | 0.89 | 2.21 | 1.81 | 25.99 | 27.78 | 32.27 | 0.610 | 1.88 | 0.148 | 0.40 | 0.17 | 0.34 |
| 48.00 | 0.84 | 1.39 | 1.39 | 26.52 | 29.07 | 33.47 | 0.602 | 1.92 | 0.164 | 0.54 | 0.21 | 0.21 |
| 48.00 | 0.94 | 1.38 | 1.38 | 26.62 | 28.98 | 33.29 | 0.640 | 1.88 | 0.177 | 0.47 | 0.19 | 0.27 |
| 50.00 | 0.55 | 1.50 | 1.50 | 28.42 | 31.43 | 36.07 | 0.676 | 2.01 | 0.199 | 0.62 | 0.26 | 0.12 |
| 50.00 | 0.60 | 1.26 | 1.26 | 28.42 | 31.15 | 35.62 | 0.579 | 2.03 | 0.149 | 0.54 | 0.23 | 0.19 |
| 50.00 | 0.70 | 2.33 | 1.02 | 28.05 | 30.68 | 34.98 | 0.545 | 1.58 | 0.176 | 0.52 | 0.22 | 0.22 |
| 50.00 | 0.79 | 2.30 | 1.40 | 27.90 | 30.39 | 34.66 | 0.574 | 1.89 | 0.154 | 0.56 | 0.23 | 0.18 |
| 50.00 | 0.84 | 1.36 | 1.36 | 27.75 | 30.28 | 34.54 | 0.506 | 1.62 | 0.155 | 0.52 | 0.22 | 0.21 |
| 52.00 | 0.50 | 1.59 | 1.59 | 30.15 | 33.28 | 37.72 | 0.653 | 2.04 | 0.201 | 0.57 | 0.29 | 0.13 |
| 52.00 | 0.55 | 1.60 | 1.60 | 29.68 | 32.77 | 37.20 | 0.645 | 2.23 | 0.204 | 0.59 | 0.28 | 0.12 |
| 52.00 | 0.60 | 1.81 | 1.81 | 29.22 | 32.39 | 36.73 | 0.688 | 2.20 | 0.199 | 0.61 | 0.27 | 0.11 |
| 52.00 | 0.65 | 2.02 | 2.01 | 28.88 | 32.16 | 36.32 | 0.660 | 2.26 | 0.263 | 0.63 | 0.26 | 0.11 |
| 52.00 | 0.84 | 2.31 | 1.53 | 28.37 | 31.54 | 35.55 | 0.631 | 2.00 | 0.153 | 0.40 | 0.23 | 0.31 |
| 54.00 | 0.79 | 1.79 | 1.79 | 29.32 | 32.81 | 36.74 | 0.730 | 1.97 | 0.194 | 0.37 | 0.27 | 0.32 |
| 54.00 | 0.84 | 2.09 | 2.09 | 29.38 | 32.69 | 36.61 | 0.658 | 2.18 | 0.248 | 0.31 | 0.27 | 0.36 |
| 54.00 | 0.94 | 1.48 | 1.47 | 29.27 | 32.47 | 36.48 | 0.655 | 1.96 | 0.195 | 0.54 | 0.27 | 0.17 |
| 56.00 | 0.65 | 2.81 | 2.46 | 30.73 | 34.60 | 38.47 | 0.667 | 2.49 | 0.277 | 0.29 | 0.29 | 0.37 |
| 56.00 | 0.74 | 2.12 | 2.12 | 30.40 | 34.16 | 38.00 | 0.649 | 2.29 | 0.260 | 0.38 | 0.29 | 0.30 |
| 56.00 | 0.89 | 2.09 | 2.09 | 29.98 | 33.80 | 37.54 | 0.661 | 2.24 | 0.249 | 0.39 | 0.29 | 0.29 |
| 56.00 | 0.94 | 1.42 | 1.42 | 30.01 | 33.71 | 37.50 | 0.644 | 1.96 | 0.185 | 0.44 | 0.27 | 0.25 |
| 56.00 | 0.99 | 2.07 | 2.07 | 29.98 | 33.74 | 37.45 | 0.621 | 2.34 | 0.268 | 0.45 | 0.30 | 0.24 |
| 58.00 | 0.55 | 2.33 | 1.69 | 32.09 | 36.61 | 40.38 | 0.646 | 2.24 | 0.186 | 0.54 | 0.33 | 0.13 |
| 58.00 | 0.65 | 2.11 | 2.11 | 31.62 | 35.81 | 39.51 | 0.642 | 2.63 | 0.340 | 0.42 | 0.31 | 0.24 |
| 58.00 | 0.74 | 2.01 | 2.01 | 31.10 | 35.41 | 38.99 | 0.666 | 2.26 | 0.268 | 0.42 | 0.32 | 0.25 |
| 58.00 | 0.94 | 2.51 | 1.99 | 30.91 | 34.88 | 38.49 | 0.656 | 2.33 | 0.248 | 0.39 | 0.30 | 0.28 |
| 58.00 | 0.99 | 1.61 | 1.61 | 30.62 | 34.84 | 38.44 | 0.679 | 2.14 | 0.229 | 0.56 | 0.31 | 0.12 |
| 60.00 | 0.50 | 2.01 | 2.01 | 33.63 | 38.53 | 41.95 | 0.626 | 2.51 | 0.293 | 0.36 | 0.32 | 0.28 |
| 60.00 | 0.60 | 2.27 | 2.27 | 32.82 | 37.38 | 40.97 | 0.604 | 2.54 | 0.294 | 0.48 | 0.33 | 0.18 |
| 60.00 | 0.94 | 2.15 | 2.15 | 31.70 | 36.04 | 39.48 | 0.602 | 2.59 | 0.305 | 0.52 | 0.33 | 0.15 |
| 62.00 | 0.50 | 2.17 | 2.17 | 34.44 | 39.75 | 42.98 | 0.603 | 2.64 | 0.297 | 0.42 | 0.32 | 0.22 |
| 62.00 | 0.60 | 2.04 | 2.04 | 33.61 | 38.62 | 42.00 | 0.658 | 2.40 | 0.282 | 0.39 | 0.33 | 0.25 |
| 62.00 | 0.70 | 2.04 | 2.04 | 33.00 | 37.92 | 41.28 | 0.633 | 2.38 | 0.276 | 0.38 | 0.33 | 0.27 |
| 64.00 | 0.74 | 2.12 | 2.12 | 33.58 | 38.90 | 42.03 | 0.607 | 2.41 | 0.278 | 0.38 | 0.33 | 0.26 |
| 64.00 | 0.84 | 2.43 | 2.43 | 33.33 | 38.53 | 41.63 | 0.589 | 2.76 | 0.352 | 0.37 | 0.33 | 0.27 |
| 66.00 | 0.60 | 2.07 | 2.07 | 35.34 | 41.02 | 43.99 | 0.627 | 2.61 | 0.343 | 0.45 | 0.36 | 0.18 |
| 66.00 | 0.94 | 2.15 | 2.15 | 33.91 | 39.46 | 42.40 | 0.609 | 2.43 | 0.315 | 0.38 | 0.33 | 0.26 |
| 68.00 | 0.65 | 2.17 | 2.17 | 35.69 | 41.80 | 44.55 | 0.598 | 2.58 | 0.318 | 0.40 | 0.35 | 0.23 |
| 70.00 | 0.60 | 2.14 | 2.14 | 36.93 | 43.47 | 45.97 | 0.606 | 2.60 | 0.294 | 0.45 | 0.37 | 0.17 |